GROVE: Reusing Solution Sketches for Current-Flow Group Selection

(With Proof Appendices)

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***Abstract*—Current-flow group closeness centrality (CFCC) selects network anchors by minimizing the grounded Laplacian inverse trace. Greedy optimization repeatedly estimates every remaining vertex's marginal contribution, requiring expensive solves after each selection. We present GROVE, which reuses two solution sketches through rank-one grounding updates. Unrefreshed candidates use the current sketches, while refreshed candidates retain inverse moments obtained through individual numerical solves. A per-round cap limits additional candidate refreshes. A uniform concentration argument, combined with residual bounds, yields a $(1-1/e-\varepsilon)$ approximation to optimal gain relative to a fixed seed under explicit sufficient conditions, whose sketch dimension is far above the empirical $d = 64$; $\varepsilon > 0$ is the accuracy parameter. On six graphs with completed comparisons, GROVE is 1.1–5.7× faster than ApproxGreedy in the main runs, and 1.4–3.8× faster in four-seed means on four graphs. Final inverse traces differ by at most 1.6% across the compared methods; the forest methods reach lower traces on roadNet-PA, and the smallest-eigenvalue heuristic of Zhou et al. is faster on DBLP. Runtime components and ablations show where reuse saves work and where updates or refreshes remain costly. On LiveJournal, with 4.0 million vertices, GROVE finishes in 4626 seconds. Supporting structural results cover completely monotone grounding objectives, and inverse-power experiments characterize the tradeoff between aggregate resistance and slow spectral modes.**

***Index Terms*—Group centrality, node grounding, completely monotone functions, supermodular optimization, randomized sketches, incremental computation.**

I. Introduction

Resource placement and leader selection [23]–[25] require groups of vertices whose collective influence depends on paths throughout a network. Current-flow group closeness centrality (CFCC) measures this influence through effective resistance [1]. Let $G = (V,E)$ be a connected undirected graph with $n = |V|$ vertices and Laplacian L. For a nonempty anchor set $S \subseteq V$, let $L_{-S} = L[V \setminus S, V \setminus S]$ be its grounded principal submatrix. Maximizing $\mathrm{CFCC}(S) = n/\operatorname{tr}(L_{-S}^{-1})$ is equivalent to minimizing the total effective resistance to the anchors.

The computational bottleneck lies in choosing each additional anchor. Greedy selection compares how much grounding each remaining vertex decreases the inverse trace, and these marginal gains change after every selection. ApproxGreedy estimates them through random projections and Laplacian solves [1]; forest-based methods use sampled spanning forests [5]. Although each round adds only one vertex, both repeat all-candidate estimation for the updated grounded system. The central question is how to reuse this work without letting stale estimates misdirect later choices.

Two natural approaches expose the difficulty. Rebuilding solution sketches every round estimates the current matrix but repeats expensive batches of solves. Alternatively, a scalar cache estimates each candidate's moments—its inverse diagonal and squared inverse-column norm—once, then subtracts algebraic decrements after each selection. The initial sketch error persists in absolute terms. As true moments shrink, their relative errors grow and may distort the ranking, and the cache gives no way to tell beforehand on which inputs this happens. The shortcut is competitive on most graphs we test and fails on one.

GROVE combines two forms of reuse. It maintains two solution sketches of the current grounded inverse, updating them through rank-one identities rather than rebuilding them through solves. Unrefreshed candidates use these current sketches. Refreshed candidates instead retain moments obtained from individual numerical solves. Unlike the scalar-cache shortcut, this second path starts from a solved column rather than a noisy initial sketch: exact decrements preserve exact moments in exact arithmetic, and residual bounds control the numerical implementation. A per-round refresh cap limits additional candidate solves. The resulting algorithm maintains current estimates while preserving work already spent on accurate candidates.

Reuse creates an adaptive estimation problem: selected vertices depend on the same randomness used in subsequent rounds. We cover all feasible grounding sets in one concentration event and combine it with numerical-error bounds. For a fixed seed S_0, k additional anchors, and accuracy parameter $\varepsilon > 0$, sufficient sketch and residual conditions yield a $(1-1/e-\varepsilon)$ approximation to the optimal gain relative to S_0. These are conservative worst-case conditions; the practical sketch dimension of $d = 64$ is evaluated empirically. Our work bound separately accounts for sketch updates and candidate refreshes.

We evaluate GROVE on seven SNAP networks [15] against ApproxGreedy, our implementations of ForestCFCM and Schur-CFCM, and Zhou's smallest-eigenvalue heuristic [6]. Matched comparisons, runtime components, ablations, parameter sweeps, and additional random seeds test selection cost and quality. The results show substantial savings on several graphs, a smaller advantage on web-Google, and small quality tradeoffs rather than uniform dominance. The largest completed input has about four million vertices.

As a supporting extension, we study completely monotone (CM) trace objectives. Hard-CFCC supermodularity and hardness are already known [1]; our structural result covers all inverse powers and heat traces under grounding. Increasing the inverse exponent emphasizes slower modes. Separate exact-marginal experiments use this parameter to trade aggregate resistance against the smallest grounded eigenvalue. These objective choices complement the CFCC algorithm.

Contributions.

1) **Hybrid CFCC selection.** Maintained solution sketches, retained refreshed moments, and capped refresh reduce repeated marginal-estimation work.
2) **Approximation and work.** A uniform adaptive event and numerical-error budgets give a conditional seeded-gain guarantee, with explicit refresh-sensitive costs.
3) **Evaluation and objective extensions.** Same-machine comparisons and component measurements assess the reuse mechanism; CM grounding structure and integer-power experiments explain the separate spectral-weighting option.

II. Problem and Preliminaries

Table I collects shared notation.

A. Grounding queries

Let $G=(V,E,w)$ be an undirected weighted graph with $n=|V|$ vertices and $m=|E|$ edges. Its nonnegative weights satisfy $w_{ij}=w_{ji}$, with $w_{ij}=0$ for nonedges. The Laplacian has entries $L_{ii}=\sum_j w_{ij}$ and $L_{ij}=-w_{ij}$ for $i\neq j$. Write $M=L+\sigma I$, where I is the identity and $\sigma\geq 0$ is a uniform regularization parameter. For a selected set $S\subseteq V$, let $\mathbf{1}_S$ be its indicator vector and $\operatorname{diag}(\mathbf{1}_S)$ the matrix with this vector on the diagonal. A grounding penalty $\beta\in(0,\infty]$ defines

$$A_S^\beta=\begin{cases}M+\beta\operatorname{diag}(\mathbf{1}_S), & \beta<\infty,\\ M[V\setminus S, V\setminus S], & \beta=\infty.\end{cases} \tag{1}$$

Finite β penalizes the selected vertices (soft grounding); hard grounding removes their rows and columns. Write $X_S=A_S^\infty$ for this active principal submatrix. When $\sigma=0$, assume connectivity and a nonempty fixed seed S_0; when $\sigma>0$, an empty seed is admissible. We consider supersets of S_0, whose nonempty grounded matrices are positive definite. The trace of a function of the empty matrix is defined as zero.

A finite-valued function $f:(0,\infty)\to[0,\infty)$ is *completely monotone* (CM) if it admits the representation

$$f(x)=\int_{[0,\infty)} e^{-tx}\,\mu(dt),\qquad \mu\geq 0, \tag{2}$$

where μ is a nonnegative measure on $[0,\infty)$ and t is the integration variable. The matrix function applies f to each eigenvalue. The algorithmic target is hard CFCC: $\beta=\infty$, $\sigma=0$, and $f(x)=x^{-1}$; the other choices support the structural extension. Starting from S_0, we select k additional anchors:

$$\min_{T\subseteq V\setminus S_0,\ |T|=k}\Phi_f(S_0\cup T),\qquad \Phi_f(S)=\operatorname{tr} f(A_S^\beta). \tag{3}$$

The *seeded gain*, normalized relative to S_0, is $F_f(T)=\Phi_f(S_0)-\Phi_f(S_0\cup T)$; its maximum over feasible k-element extensions is OPT_k. All approximation factors below compare extensions of this same seed. The experimental total budget is $K=|S_0|+k$: one seed and $K=20$ total anchors mean $k=19$ additions.

For a candidate $v\notin S$, its marginal cost reduction is $\Delta_f(v\mid S)=\Phi_f(S)-\Phi_f(S\cup\{v\})$. Supermodularity means $\Delta_f(v\mid S)\geq\Delta_f(v\mid U)$ whenever $S\subseteq U$ and $v\notin U$. Thus the seeded gain is submodular; when it is also monotone, exact greedy selection achieves the standard $1-1/e$ approximation to OPT_k [10].

B. Current flow and spectral weighting

For hard grounding with $\sigma=0$, injecting one unit of current at an ungrounded vertex u produces voltage $(L_{-S}^{-1})_{uu}$ at u, equal to its effective resistance to the grounded set. Summing these resistances gives the denominator of CFCC [1], so minimizing $\Phi_1(S)=\operatorname{tr}L_{-S}^{-1}$ maximizes CFCC.

More generally, inverse powers $f(x)=x^{-s}$, $s>0$, give $\Phi_s(S)=\operatorname{tr}(A_S^\beta)^{-s}$. If λ_i are the positive eigenvalues of A_S^β, then $\Phi_s(S)=\sum_i\lambda_i^{-s}$. For $\lambda_i<\lambda_j$, their relative weights are $(\lambda_j/\lambda_i)^s$; increasing s therefore emphasizes slower spectral modes. Heat instead weights each mode by $e^{-\tau\lambda_i}$ at horizon $\tau>0$. CFCC is our main computational and experimental objective. The broader family provides interpretable spectral weights.

The hard-grounding CFCC cost Φ_1 is nonincreasing and supermodular, and exact group selection is NP-hard [1]. These known properties are all that the greedy algorithm and its analysis in Sections III and IV require; Section V extends them to the CM family.

III. The Hybrid CFCC Algorithm

We now specialize to hard grounding with $\sigma=0$ and CFCC ($s=1$). To avoid rebuilding all marginal estimates after every anchor, GROVE maintains solution sketches for unrefreshed candidates and scalar moments for candidates already refreshed by a solve. The two paths preserve different information about the current matrix (Fig. 1(b)).

Let $W=V\setminus S_0$ contain the $N=n-|S_0|$ initial candidates. The seed matrix is $A_0=L[W,W]=C^\top C\in\mathbb{R}^{N\times N}$, where $C\in\mathbb{R}^{m\times N}$ is the oriented edge-incidence matrix, scaled by square roots of edge weights and restricted to columns in W. This exact incidence factor is distinct from the approximate Cholesky factor used for preconditioning. For additional anchors $T\subseteq W$, let R_T be the current active inverse, embedded in $\mathbb{R}^{N\times N}$ with zero rows and columns at T. Write $R=R_T$, let e_u be the

TABLE I
NOTATION SHARED ACROSS SECTIONS. AUXILIARY PROOF VARIABLES ARE LOCAL TO THEIR STATEMENTS.

Symbol	Meaning	Symbol	Meaning
L, M	Laplacian and $M = L + \sigma I$	s, p	Real inverse exponent and positive integer exponent
S_0, N	Fixed seed and $N = n - \lvert S_0 \rvert$ initial candidates	Φ_s^0	Initial inverse-power cost $\Phi_s(S_0)$
T, k, K	Added anchors, additional budget, total budget	$\Omega(u,v)$	Pairwise effective resistance
A_0, C, R_T	Seed matrix, incidence factor, current padded inverse	Y_1, Y_2	CFCC solution sketches for first/second moments
Π_C, Π_I	Independent random projection matrices	$\varepsilon, \varepsilon_{\text{forest}}$	GROVE accuracy parameter and forest-method input
$e_u, r_j(u,T)$	Coordinate vector and $e_u^\top R_T^j e_u$	$\bar{R}_v, s_x$	Inverse after selecting v, and $\lVert Re_v \rVert^2$
$d, B_{\text{init}}, B_r, b_r$	Sketch rows, initial/refresh batch widths, refresh cap	θ, ζ, δ	Acceptance, moment-error, and failure tolerances
$n_{\text{ref}}, n_t^{\text{ref}}$	Distinct refreshed candidates in total/in round t	$\alpha_{\text{step}}, \alpha_{\text{cost}}$	Greedy-step fraction and trace-cost factor
B_0, B_U	Seed approximate inverse and active restriction	$\mathscr{E}, \rho_{\text{res}}$	Sketch-error matrix and column-residual bound
$F_c, \mathscr{T}_t$	Factor entries and work of a solve in round t	$z_{\text{ser}}, \vartheta$	Formal-series variable and removal interpolation

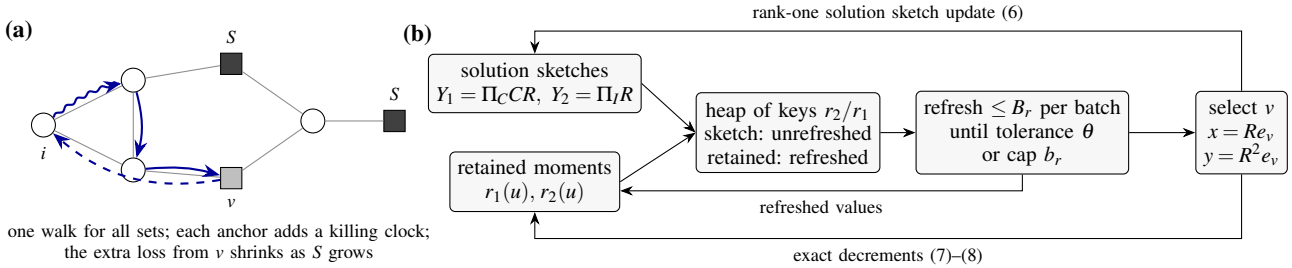


Fig. 1. (a) The CM coupling: all grounding sets share one walk and one family of killing clocks, so a new anchor removes fewer surviving returns when more anchors are present. (b) The maintained state of GROVE: unrefreshed candidates are keyed from the current solution sketches, refreshed candidates from retained exact moments; after each selection both paths are updated from the selected inverse columns.

coordinate unit vector at an active candidate u, and define $r_j(u,T) = e_u^\top R_T^j e_u$ for integer $j \ge 1$. The CFCC marginal is

$$\Delta_1(u \mid S_0 \cup T) = \frac{r_2(u,T)}{r_1(u,T)}, \qquad r_1 = R_{uu}, \quad r_2 = \lVert Re_u \rVert^2. \quad (4)$$

Here $\lVert \cdot \rVert$ is the Euclidean norm. Both moments come from one inverse column, so a candidate refresh requires one solve rather than two complete trace estimates.

A. Common-seed initialization

The initial anchor approximately minimizes L^+_{vv}, where L^+ is the Moore–Penrose pseudoinverse. Writing $\Omega(u,v)$ for pairwise effective resistance, the identity $\sum_u \Omega(u,v) = \operatorname{tr} L^+ + nL^+_{vv}$ gives the singleton CFCC rule [1], [5]. We estimate this diagonal with an incidence-based random-sign sketch, projecting right-hand sides and solutions onto $\mathbf{1}^\perp = \{x : \mathbf{1}^\top x = 0\}$, where $\mathbf{1}$ is the all-ones vector. Every compared method receives the common seed $S_0 = \{s_0\}$. Its computation is a separate shared stage; all later gains use this seed as their fixed baseline.

Initializing scalar estimates once saves solves, but subtracting exact decrements preserves their initial absolute error. As moments shrink, this error can distort their ratios. GROVE instead maintains sketches of the *current* inverse. Draw independent random matrices $\Pi_C \in \mathbb{R}^{d\times m}$ and $\Pi_I \in \mathbb{R}^{d\times N}$ with independent mean-zero entries of variance $1/d$. Starting at $R = A_0^{-1}$, construct the two $d \times N$ solution sketches

$$Y_1 = \Pi_C CR, \qquad Y_2 = \Pi_I R, \qquad \widehat{r}_j(u) = \lVert Y_j[:,u] \rVert^2 \quad (j = 1,2). \quad (5)$$

The notation $Y_j[:,u]$ denotes column u. The identity $RA_0R = R$ gives $\lVert CRe_u \rVert^2 = r_1(u)$ and $\lVert Re_u \rVert^2 = r_2(u)$, so the squared column norms estimate the two current moments. Initialization uses d sparse-system right-hand sides per solution sketch.

B. Two complementary update paths

Grounding one vertex gives a rank-one inverse correction, which can be applied directly to both solution sketches. An *unrefreshed* candidate uses the updated solution-sketch column norms. A *refreshed* candidate has undergone an individual solve and thereafter retains its moments through scalar updates. The following identities describe both paths in exact arithmetic.

Proposition 1 (Hybrid state identities)**.** *On selecting v, let $x = Re_v$, $y = R^2 e_v$, $c = 1/x_v$, and $s_x = \lVert x \rVert^2$. The next inverse is $R' = R - cxx^\top$, with a zero row and column at v. Both solution sketches update as*

$$Y_j' = Y_j - cY_j[:,v]x^\top, \qquad j = 1,2. \quad (6)$$

For each remaining refreshed candidate u, retain its moments by

$$r_1'(u) = r_1(u) - cx_u^2, \quad (7)$$

$$r_2'(u) = r_2(u) - 2cx_u y_u + c^2 s_x x_u^2. \quad (8)$$

With exact columns, exact refreshed moments remain exact. A scalar cache initialized with error $e_j(u)$ retains that signed absolute error under exact independent decrements. The solution sketch update costs $O(Nd)$ work and storage; scalar retention costs $O(N)$ work, once x, y are available.

Proof sketch. The principal-inverse identity gives $R' = R - cxx^\top$. Multiplication by the fixed sketch factors gives (6). Taking diagonals gives (7); squaring first gives (8). Subtracting

Algorithm 1 GROVE: hard-CFCC hybrid selection

Input: G, common seed s_0, total budget K; $d, B_{\text{init}}, B_r, b_r, \theta$; solve tolerances $\varepsilon_{\text{init}}, \varepsilon_{\text{ref}}, \varepsilon_{\text{upd}}$.

1. Ground s_0; construct the seed preconditioner if enabled. Initialize Y_1, Y_2 and moments using batches of width B_{init} and tolerance $\varepsilon_{\text{init}}$. Mark all candidates unrefreshed.
2. For rounds $1, \dots, K-1$:
   a. Build heap $\mathscr{H}$ with key $r_2(u)/r_1(u)$. Set incumbent value $b = -\infty$, winner $v = \perp$, refresh count $r = 0$.
   b. Consume refreshed heap tops and update (v, b) from their values. Let q_{heap} be the largest remaining key, or $-\infty$ if empty.
   c. If $v \neq \perp$ and $b \geq (1-\theta) q_{\text{heap}}$, stop this round. If $v \neq \perp$, $b_r > 0$, and $r \geq b_r$, also stop.
   d. Pop entries until up to B_r unrefreshed candidates are collected; refreshed entries encountered update the incumbent without a solve. Refresh the collected candidates at ε_{ref}, mark them refreshed, update (v, b), and retain a newly winning column Re_v. Increase r by the batch size and return to step 2b.
   e. Before a nonfinal round ends, obtain $x = Re_v$ and $y = R^2 e_v$. Reuse a retained winning column when enabled; solve any missing columns at ε_{upd}. Update both solution sketches by (6). For unrefreshed candidates recompute moment norms; for refreshed candidates apply (7)–(8).
   f. Zero coordinate v, add v to the grounded set, and mask it in subsequent solves. Skip maintenance after the final selection.
3. Return the K grounded nodes, including s_0.

the same decrement from a true moment and its cached estimate preserves their signed difference. □

After each selection, GROVE updates both solution sketches for all remaining vertices and the scalar moments of refreshed candidates. Retention allows a later round to choose an already refreshed candidate without another refresh. A winning refresh provides x; the additional column y supports scalar maintenance. Both are reused when available. Numerical solves introduce errors controlled by the tolerances below and analyzed in Section IV.

C. Capped candidate refresh

At each round, rebuild a max-heap from the current moment ratios, using solution sketch estimates for unrefreshed candidates and retained values for refreshed ones. Refreshed heap tops update the incumbent without a solve; unrefreshed candidates are refreshed in blocks of at most B_r. Let b be the best refreshed marginal found and q_{heap} the largest remaining key. Stop when $b \geq (1-\theta) q_{\text{heap}}$, with acceptance tolerance $0 \leq \theta < 1$, or when the positive refresh cap b_r is reached. A zero cap means unlimited refreshes. The cap controls candidate solves, not solution sketch maintenance. It is checked between batches, so a positive cap can be exceeded by fewer than B_r refreshes; the initial maximum-key candidate is always processed before a cap stop.

Algorithm 1 uses total budget K, initialization batch width B_{init}, and refresh batch width B_r. The relative solve tolerances $\varepsilon_{\text{init}}, \varepsilon_{\text{ref}}, \varepsilon_{\text{upd}}$ apply to initialization, candidate refreshes, and newly computed update columns, respectively. A reused winning column retains its ε_{ref} tolerance. The seed preconditioner is constructed once and masked for later solves.

Classical lazy greedy (CELF) reuses stale exact marginal upper bounds from submodularity [11]. A raw JL estimate is not such an upper bound: combining ApproxGreedy with lazy ordering requires calibrated intervals or fresh accuracy events. GROVE instead updates current estimates and retains individually refreshed moments; its cap is analyzed on the uniform event in Section IV. Two restrictions isolate the update paths in Section VI. *Sketch-only* maintains both solution sketches but rekeys every candidate from them each round, removing the retained scalar path and its y solve. *Scalar cache* initializes moments once and applies algebraic decrements without maintaining solution sketches. Their comparison tests whether current sketches and retained values reduce different sources of repeated work.

IV. Approximation and Work

Reuse saves solves but couples later choices to the original random sketches. We first control this adaptivity, then incorporate numerical errors, summarize the solver policy, and account for the work that remains after eliminating repeated all-candidate solves.

A. One event for adaptive selection

Accuracy for a fixed matrix is insufficient when the selected anchors depend on the sketches. We therefore cover every feasible grounding set in one event. Let δ be its failure probability and ζ the relative moment-error tolerance. The count $\mathscr{N}_{N,k}$ below includes every active candidate paired with a set of at most k added anchors; α_{step} is the resulting per-round greedy factor.

Theorem 2 (Adaptive hybrid guarantee)**.** *Fix S_0, $1 \leq k < N$ additions, $0 < \delta, \zeta < 1$, and define*

$$\mathscr{N}_{N,k} = \sum_{t=0}^{k} \binom{N}{t} (N-t), \qquad d \geq \left\lceil 8\zeta^{-2} \log \frac{4\mathscr{N}_{N,k}}{\delta} \right\rceil.$$

Use independent Gaussian entries of variance $1/d$, or independent signs $\pm 1/\sqrt{d}$, in the two solution sketches. With probability at least $1-\delta$, exact maintenance gives relative error at most ζ for both moments at every active vertex of every $|T| \leq k$. For exact refreshed values, put $\chi = (1+\zeta)/(1-\zeta)$ and

$$\alpha_{\text{step}} = \min\{\chi^{-2}, (1-\theta)\chi^{-1}\}.$$

Algorithm 1, with any positive cap or unlimited refreshes, returns gain at least $(1-e^{-\alpha_{\text{step}}})\text{OPT}_k$. In particular, $\zeta \leq \varepsilon/8$ and $\theta \leq \varepsilon/2$ give $(1-1/e-\varepsilon)\text{OPT}_k$. The sufficient dimension is $O(\varepsilon^{-2}((k+1)\log(N+1) + \log(1/\delta)))$.

Proof sketch. One fixed column. For a deterministic nonzero target vector w and one unscaled random sketch row g, the normalized projection $Z = g^\top w/\|w\|$ satisfies $\mathbb{E}Z^2 = 1$ and $\mathbb{E}e^{tZ} \leq e^{t^2/2}$ for both row distributions, hence $\mathbb{E}e^{tZ^2} \leq (1-2t)^{-1/2}$ for $0 < t < 1/2$. A Chernoff bound on the average of d copies of Z^2 gives upper-tail failure at most $e^{-d\zeta^2/8}$; the inequality $e^{-y} \leq 1 - y + y^2/2$ and $\mathbb{E}Z^4 \leq 3$ give lower-tail failure at most $e^{-d\zeta^2/6}$.

All adaptive trajectories. The targets $CR_T e_u$ and $R_T e_u$, over all $|T| \leq k$ and active u, form a *deterministic* family of $2\mathscr{N}_{N,k}$ vectors. A union bound puts all of them inside relative error

ζ with probability at least $1-\delta$. The maintained solution sketches equal $\Pi_C C R_T$ and $\Pi_I R_T$ exactly for whatever T the algorithm reaches, so adaptive selection, refresh patterns, and stopping decisions stay on this event.

One round. Write Δ_u for candidate u's true current marginal. Every key lies in $[\Delta_u/\chi, \chi\Delta_u]$, and refreshed keys are exact. Let u_* maximize Δ_u, let w carry the largest initial key, and retain the incumbent b and remaining maximum key q_{heap} from Algorithm 1. At a cap stop w has been processed, so the incumbent satisfies $b \geq \Delta_w \geq \text{key}(w)/\chi \geq \Delta_{u_*}/\chi^2$. At an acceptance stop, processed values are at most b and an unprocessed u has $\Delta_u \leq \chi q_{\text{heap}} \leq \chi b/(1-\theta)$; hence $b \geq (1-\theta)\Delta_{u_*}/\chi$.

Greedy recurrence. By monotonicity and supermodularity of the hard-CFCC cost [1], the gap to OPT_k is at most k times the largest marginal, so it contracts by $1-\alpha_{\text{step}}/k$ per round, giving $1-(1-\alpha_{\text{step}}/k)^k \geq 1-e^{-\alpha_{\text{step}}}$. Finally $1-\chi^{-2} \leq 4\zeta$, $1-(1-\theta)\chi^{-1} \leq \theta+2\zeta$, and $e^{-\alpha_{\text{step}}}-e^{-1} \leq 1-\alpha_{\text{step}}$, while $\mathcal{N}_{N,k} \leq (k+1)N^{k+1}$ yields the dimension order. □

The sufficient dimension records the price of adaptive reuse. Fresh independent solution sketches require $d_{\text{fresh}} = \lceil 8\zeta^{-2}\log(4Nk/\delta)\rceil$ for the corresponding ideal-solve event: each current matrix is fixed conditional on preceding rounds. Maintained solution sketches instead cover all feasible matrices at initialization. This extra dependence on k trades a larger uniform-accuracy requirement for fewer repeated solves.

The bound is a worst-case sufficient condition. For example, $\varepsilon = 0.5$, $\zeta = \varepsilon/8$, $\delta = 0.1$, and $k = 19$ give $d \geq 445{,}823$ at $N = 317{,}079$ and $d \geq 492{,}870$ at $N = 10^6$. Our experimental $d = 64$ is an empirical operating point, outside these sufficient-dimension settings; Table VIII evaluates its accuracy–work tradeoff.

B. Numerical accuracy

Practical solves perturb both update paths. For a computed solution sketch $\widetilde{Y}$ and its ideal counterpart Y, write $\mathscr{E} = \widetilde{Y} - Y$ and measure the error by $\|\mathscr{E}\|_{F,A_0} = \sqrt{\text{tr}(\mathscr{E}A_0\mathscr{E}^\top)}$. Let e_1, e_2 bound this norm for the first and second solution sketches. Let $a > 0$ and $B > 0$ bound the smallest and largest eigenvalues of A_0, respectively. The next result propagates a selected-column residual into solution sketch and moment errors. Supported errors and columns have zero coordinates at already grounded vertices.

Proposition 3 (Residual-controlled maintenance)**.** *Let $a \leq \lambda_{\min}(A_0)$ and $B \geq \lambda_{\max}(A_0)$. Suppose a supported selected column $\widetilde{x}$ has $\widetilde{x}_v > 0$ and true active-system residual norm at most ρ_{res}. If a solution sketch's current error is at most e in $\|\mathscr{E}\|^2_{F,A_0} = \text{tr}(\mathscr{E}A_0\mathscr{E}^\top)$, its update has error at most*

$$e^+ = e + \frac{\|\widetilde{Y}[:,v]\|\rho_{\text{res}}}{\sqrt{a}\,\widetilde{x}_v} + \omega, \tag{9}$$

where ω bounds supported rounding error in the same norm. On the ideal-sketch event with tolerance ζ_0, the relative moment errors are bounded by

$$\zeta_0 + 2\sqrt{1+\zeta_0}\,e_1 + e_1^2, \quad \zeta_0 + 2\sqrt{1+\zeta_0}\sqrt{B}\,e_2 + Be_2^2.$$

Thus a target tolerance $\bar{\zeta} < 1$ is met by $\zeta_0 = \bar{\zeta}/2$ and $e_1, \sqrt{B}e_2 \leq \bar{\zeta}/8$.

Proof sketch. Let U be the active coordinates, $x = R_T e_v$, $h = \widetilde{x} - x$, and $\|w\|^2_{A_0} = w^\top A_0 w$. The residual gives $\|h\|_{A_0} \leq \rho_{\text{res}}/\sqrt{a}$. Active inverse equations give $\langle h, x\rangle_{A_0} = h_v$ and $\|x\|^2_{A_0} = x_v$; hence

$$\left\| \frac{\widetilde{x}}{\widetilde{x}_v} - \frac{x}{x_v} \right\|^2_{A_0} = \frac{\|h\|^2_{A_0} - h_v^2/x_v}{\widetilde{x}_v^2} \leq \frac{\rho_{\text{res}}^2}{a\widetilde{x}_v^2}.$$

Put $P = I - e_v x^\top/x_v$ and $\mathscr{E} = \widetilde{Y} - Y$. Expanding the supported-row energies yields

$$\|\mathscr{E}P\|^2_{F,A_0} = \|\mathscr{E}\|^2_{F,A_0} - \|\mathscr{E}[:,v]\|^2/x_v.$$

The computed-minus-ideal update is

$$\widetilde{Y}' - Y' = \mathscr{E}P + D_{\text{rnd}} + \widetilde{Y}[:,v](x/x_v - \widetilde{x}/\widetilde{x}_v)^\top,$$

where $\|D_{\text{rnd}}\|_{F,A_0} \leq \omega$. Contraction of $\mathscr{E}P$, the outer-product norm, and the triangle inequality give (9). For sketch j, the coordinate-representer bound is $\|\mathscr{E}[:,u]\| \leq e_j\sqrt{r_1(u)}$. For Y_1 this is relative to $\sqrt{r_1}$; for Y_2 use $r_1 \leq Br_2$. Expanding the perturbed squared norms on the ideal-sketch event gives the two stated moment-error bounds. □

Retained scalar moments follow a separate error path: their refresh error plus accumulated decrement-error bounds. The coefficients use independently solved selected columns, so candidate-cache errors do not feed back into them. To combine both paths, let $\nu < 1$ bound the relative error of computed refreshed marginals and let $\bar{\zeta} < 1$ be the target solution sketch-moment tolerance. Define

$$\eta = \max\left\{ \frac{1+\bar{\zeta}}{1-\bar{\zeta}} - 1, \nu \right\} < 1.$$

A sufficient inexact step factor is

$$\alpha_{\text{num}} = \frac{1-\nu}{1+\nu}\min\left\{ \frac{1-\eta}{1+\eta}, (1-\theta)(1-\eta) \right\}.$$

The minimum follows the two stopping cases in Theorem 2; the prefactor accounts for choosing the incumbent using approximate refreshed values. Assigning sampling and numerical budgets so that $\alpha_{\text{num}} \geq 1-\varepsilon$ gives the stated $(1-1/e-\varepsilon)$ seeded-gain factor. Thus sketch concentration, column residuals, and retained-moment errors enter one approximation condition.

C. Solver implementation

GROVE constructs an approximate-Cholesky seed preconditioner and masks grounded coordinates in later applications. Blocks with fewer than four right-hand sides use Jacobi preconditioning; on roadNet-PA the seed factor is used for every block. This mixed solver policy is part of the evaluated implementation.

For work accounting, suppose the seed approximate inverse B_0 satisfies $a_c A_0^{-1} \preceq B_0 \preceq b_c A_0^{-1}$, with $0 < a_c \leq b_c$, and put $\chi_c = b_c/a_c$. Here $P \preceq Q$ means $Q - P$ is positive semidefinite.

For active coordinates U after t additions, define $A = A_0[U,U]$ and $B_U = B_0[U,U]$. The preconditioned conjugate-gradient (PCG) error is measured in $\|e_h\|_A = \sqrt{e_h^\top A e_h}$, where e_h is the solution error after h iterations. Under the stated seed-quality premise, $t + O(\sqrt{\chi_c}\log(1/\xi))$ iterations suffice for relative energy accuracy ξ, capped by $|U|$. The block inverse formula gives $(A_0^{-1})[U,U] = A^{-1} + ZZ^\top$, where the positive Schur-complement correction has $\operatorname{rank} Z \le t$. After congruence by $A^{1/2}$, restricting the seed bounds shows that $A^{1/2}B_U A^{1/2}$ has no eigenvalue below a_c and at most t above b_c. A PCG residual polynomial places roots at the large outliers and uses a Chebyshev factor on $[a_c, b_c]$. The outlier factors have magnitude at most one on that interval. For $\chi_c > 1$ and $h \ge t$, this gives

$$\frac{\|e_h\|_A}{\|e_0\|_A} \le 2\left(\frac{\sqrt{\chi_c}-1}{\sqrt{\chi_c}+1}\right)^{h-t},$$

which yields the iteration bound. For $\chi_c = 1$, at most $t+1$ iterations suffice; the dimension cap is $|U|$ in either case. The Kyng–Sachdeva construction [3] supplies one admissible seed-quality guarantee. The implementation uses a lightweight clique-sampling variant, with separate condition-number-dependent bounds for its Jacobi calls.

D. Work and comparison with fresh-round selection

The work saved by reuse depends on how often solution sketch updates and retained values replace candidate solves. Let T_{build} be preconditioner construction work, n_{ref} the total number of distinct refreshed candidates, n_t^{ref} the count in round t, F_c the number of stored factor entries, and $\mathscr{T}_t$ the work of one solve at the required precision in that round. The hybrid's work is

$$T_{\text{build}} + O\Bigg(dm + 2d\mathscr{T}_0 + \sum_{t=0}^{k-1} n_t^{\text{ref}}\mathscr{T}_t + 2\sum_{t=0}^{k-2}\mathscr{T}_t + kNd + kN + (kn_{\text{ref}} + n_{\text{ref}} + k)\log(N+1)\Bigg). \tag{10}$$

The dm term forms the incidence-based sketch right-hand sides, and $2d\mathscr{T}_0$ covers initial solution sketch solves; the two sums cover candidate refreshes and selected-column maintenance. Reusing the winning column reduces the latter. The kNd and kN terms cover solution sketch and scalar updates, respectively. The solution sketches require $O(Nd)$ storage in addition to the graph, factor, scalar moments, and solve batches. Exact retention makes $n_{\text{ref}} \le N$; with a common positive cap and batch width, $n_{\text{ref}} \le \min\{N, k(b_r + B_r - 1)\}$. The heap term covers previously refreshed candidates that are processed again without a solve.

Under the preceding seed-quality premise, an all-factor configuration can take

$$\mathscr{T}_t = O\big((m+N+F_c)\min\{N-t, t+1+\sqrt{\chi_c}\log(2/\xi)\}\big).$$

Jacobi calls instead use their own condition-number-dependent iteration bound. The maintained algorithm is nearly linear in graph size for fixed budget, precision, and refresh parameters. Substituting the uniform dimension into (10) exposes an array-update term of order $\widetilde{O}(k^2 N\varepsilon^{-2})$, alongside initialization and selected/refreshed solves.

At $s = 1$, fresh-round JL selection with an SDDM solver has $\widetilde{O}(km\varepsilon^{-2})$ selection work, as in ApproxGreedy [1], [12]. This uses fresh $O(\varepsilon^{-2}\log(Nk/\delta))$-row sketches and accurate solves in each round. In contrast, (10) retains the maintained method's explicit budget, array-update, and refresh costs; empirical savings arise when updates replace enough solves.

V. Extension: Completely Monotone Objectives

The CFCC algorithm uses only $s = 1$ and the known hard-grounding structure [1]. This section extends that structure to every CM trace objective and supports the separate inverse-power experiment in Section VI-G. Positive integer powers $p \ge 1$ provide a tunable objective for exact-marginal selection on medium graphs, allowing aggregate resistance and slow modes to receive different weights.

A. CM grounding structure

The *admissible lattice* is the family of sets containing the fixed seed S_0; an *available vertex* lies outside the current selected set.

Theorem 4 (CM grounding structure)**.** *For every CM function f and every $\beta \in (0,\infty]$, Φ_f is nonincreasing and supermodular on the admissible lattice. At an available vertex, a soft-grounding drop is strictly positive exactly when f is nonconstant; a hard-grounding drop is strictly positive exactly when $f \not\equiv 0$. Writing $a_f = \lim_{x\to\infty} f(x) = \mu(\{0\})$, the limit is*

$$\lim_{\beta\to\infty} \operatorname{tr} f(A_S^\beta) = \operatorname{tr} f(A_S^\infty) + |S| a_f.$$

In particular, inverse powers and heat with $\tau > 0$ converge directly to the hard-grounding objective.

Proof sketch. Couple all sets through one continuous-time walk X_t with jump rates w_{ij} (Fig. 1(a)). For finite β, draw independent exponential clocks E_0, E_v of rates σ, β, taking $E_0 = \infty$ if $\sigma = 0$. Define occupation time $\ell_v(t) = \int_0^t \mathbf{1}\{\mathsf{X}_u = v\}\,du$, $\kappa_v = \inf\{t : \ell_v(t) \ge E_v\}$, and killing time $\mathsf{T}_S = \min\{E_0, \min_{v\in S}\kappa_v\}$. For hard grounding, κ_v is the first hitting time of v. Then $\mathsf{T}_{S\cup T} = \min\{\mathsf{T}_S, \mathsf{T}_T\}$ pathwise. Conditional soft survival is $e^{-\sigma t - \beta\sum_{v\in S}\ell_v(t)}$, giving generator $-A_S^\beta$. Hard absorption gives the principal-matrix semigroup, embedded by zeros on S.

For a walk starting at i, put $Z_S = \mathbf{1}\{\mathsf{X}_t = i,\ t < \mathsf{T}_S\}$. If $S \subseteq T$ and $v \notin T$, then $Z_S \ge Z_T$ and

$$\begin{aligned} Z_S - Z_{S\cup\{v\}} &= Z_S\mathbf{1}\{\kappa_v \le t\} \\ &\ge Z_T\mathbf{1}\{\kappa_v \le t\} = Z_T - Z_{T\cup\{v\}}. \end{aligned}$$

Expectations summed over i give the claims for $h_S^\beta(t) = \operatorname{tr} e^{-tA_S^\beta}$. The finite traces and Tonelli's theorem let these nonnegative drops be integrated against μ in (2).

For strictness, the no-jump event from v contributes $e^{-(d_v+\sigma)t}(1-e^{-\beta t})$ to a soft drop and $e^{-(d_v+\sigma)t}$ to a hard drop, where $d_v = \sum_j w_{vj}$. The hard drop at $t = 0$ is one. Thus the soft

case is strict exactly when $\mu((0,\infty)) > 0$, or f is nonconstant; the hard case is strict exactly when $\mu \neq 0$.

For the limit, use common unit-rate clocks and set $E_v^\beta = E_v^1/\beta$. At every $t > 0$, survival converges almost surely to hard survival, so bounded convergence gives $h_S^\beta(t) \to h_S^\infty(t)$. For fixed $\beta_0 > 0$, $h_S^{\beta_0}$ is an integrable upper bound for $\beta \geq \beta_0$. At $t = 0$ the soft and hard traces are n and $n - |S|$, respectively. Dominated convergence therefore gives the stated limit, including the atom $|S|\mu(\{0\})$. □

B. Integer marginal evaluation

For an active inverse R and candidate v, put $x = Re_v$, $c = 1/r_1(v)$, and $\bar{R}_v = R - cxx^\top$. The marginal $\Delta_p = \operatorname{tr} R^p - \operatorname{tr} \bar{R}_v^p$ can be evaluated from moments $r_j(v) = e_v^\top R^j e_v$, avoiding separate traces. Set $a_j = c r_{j+1}(v)$ and let g_j be the degree-j coefficient of $\log(r_1 - r_2 z_{\mathrm{ser}} + r_3 z_{\mathrm{ser}}^2 - \cdots)$, with formal series variable z_{ser}. Then $\Delta_p = (-1)^p p g_p$, so moments through $p+1$ suffice. For example,

$$\Delta_2 = 2r_3/r_1 - (r_2/r_1)^2. \tag{11}$$

The following identity exposes the nonnegative terms behind such alternating expressions.

Proposition 5 (Integer marginal identity)**.** *For $j \geq 1$, define $d_j = cx^\top \bar{R}_v^{j-1} x \geq 0$, $H_0 = 1$, and $H_j = \sum_{i=1}^j d_i H_{j-i}$. For every integer $p \geq 1$, the inverse-power marginal is*

$$\Delta_p = \operatorname{tr} R^p - \operatorname{tr} \bar{R}_v^p = \sum_{j=1}^p j d_j H_{p-j}.$$

All terms are nonnegative, and moments through $r_{p+1}(v)$ determine the marginal.

Proof. For the formal variable z_{ser}, define $\mathscr{D}(z_{\mathrm{ser}}) = \sum_{j\geq 1} d_j z_{\mathrm{ser}}^j$. Since $R = \bar{R}_v + cxx^\top$, the determinant lemma and the resolvent series give, near zero,

$$\frac{\det(I - z_{\mathrm{ser}}\bar{R}_v)}{\det(I - z_{\mathrm{ser}} R)} = \frac{1}{1 - \mathscr{D}(z_{\mathrm{ser}})},$$
$$\sum_{p\geq 1} \frac{\Delta_p}{p} z_{\mathrm{ser}}^p = -\log(1 - \mathscr{D}(z_{\mathrm{ser}})).$$

The coefficients of $H(z_{\mathrm{ser}}) = (1 - \mathscr{D}(z_{\mathrm{ser}}))^{-1}$ obey the stated recurrence. Differentiating the logarithmic identity and comparing coefficients gives $\Delta_p = \sum_{j=1}^p j d_j H_{p-j}$. Positive semidefiniteness of $\bar{R}_v$ gives $d_j, H_j \geq 0$. Applying the determinant lemma in the reverse direction also gives

$$H(z_{\mathrm{ser}}) = 1 + \sum_{j\geq 1} c r_{j+1}(v) z_{\mathrm{ser}}^j.$$

Thus coefficients through degree p require only the stated moments. □

The power experiments obtain the required moments by successive linear solves and use numerical exact-marginal selection.

C. From trace weighting to the smallest eigenvalue

For a symmetric positive-definite (SPD) grounded matrix, the inverse trace combines all eigenvalues, whereas the smallest eigenvalue describes the slowest mode. Their connection depends on the guarantee available for selection. A trace-cost factor $\alpha_{\mathrm{cost}} \geq 1$ compares final costs; a seeded-gain factor $\gamma \in [0,1]$ compares reductions from the initial cost $\Phi_s^0 = \Phi_s(S_0)$. Let $D = n - |S_0| - k$ be the final active dimension, λ_k^* the largest attainable smallest eigenvalue among extensions of this seed, and X_{out} the output's grounded matrix. The next result keeps the baseline dependence explicit.

Proposition 6 (A baseline-aware spectral bridge)**.** *For an SPD matrix X of dimension D and every $s > 0$,*

$$\lambda_{\min}(X)^{-1} \leq (\operatorname{tr} X^{-s})^{1/s} \leq D^{1/s} \lambda_{\min}(X)^{-1}.$$

At a fixed final dimension D, an α_{cost}-approximation to minimum trace, $\alpha_{\mathrm{cost}} \geq 1$, has spectral factor $(\alpha_{\mathrm{cost}} D)^{-1/s}$ relative to the largest attainable smallest eigenvalue λ_k^*. If instead the available guarantee is gain factor γ, write $\Phi_s^0 = \Phi_s(S_0)$ and $\rho_{\mathrm{gain}} = 1 - \gamma$. Then

$$\frac{\lambda_{\min}(X_{\mathrm{out}})}{\lambda_k^*} \geq [\rho_{\mathrm{gain}} \Phi_s^0 (\lambda_k^*)^s + \gamma D]^{-1/s}.$$

The inverse-power root approaches the reciprocal smallest eigenvalue uniformly, with relative excess at most $e^{\log D/s} - 1$.

Proof sketch. Factor $\lambda_{\min}^{-s}$ from the trace; the remaining D terms are at most one and at least one equals one. For the gain bound, use $\Phi_s(\mathrm{out}) \leq \rho_{\mathrm{gain}} \Phi_s^0 + \gamma \min \Phi_s$ and $\min \Phi_s \leq D(\lambda_k^*)^{-s}$. Combining the two inequalities yields the baseline-dependent spectral factor. □

The bound explains how inverse powers approach the spectral objective, while retaining the distinction between final-cost and gain guarantees. A finite grounding decision can therefore rank vertices differently from a leading-eigenvector sensitivity score. Zhou et al. [6] use the Fast score $h_v = 2\phi_1(v) \sum_{u\sim v, u\notin S} \phi_1(u)$ for unit weights, where ϕ_1 is a unit eigenvector for the smallest grounded eigenvalue and $u \sim v$ denotes adjacency. In comparison, define the finite-removal path $Q_p(\vartheta) = \operatorname{tr} R^p - \operatorname{tr}(R - \vartheta cxx^\top)^p$ for local interpolation parameter $\vartheta \in [0,1]$. The endpoints represent no removal and complete removal, respectively. Its initial sensitivity is

$$Q_p'(0) = \frac{p}{r_1(v)} \sum_i \lambda_i^{-(p+1)} \phi_i(v)^2,$$

where λ_i, ϕ_i are the current grounded eigenvalues and orthonormal eigenvectors. Thus all modes contribute, with smaller eigenvalues weighted more strongly. At $p = 2$, $Q_2(\vartheta) = 2\vartheta a_2 - \vartheta^2 a_1^2$ also includes a finite-removal correction at $\vartheta = 1$, as follows by expanding the squared updated inverse and taking its trace. These differences motivate evaluating both inverse trace and smallest eigenvalue as p varies, separately from testing the efficiency of the large-scale $p = 1$ algorithm.

VI. Experimental Evaluation

We test whether reusing current sketches and retained moments reduces selection cost while preserving CFCC quality. The comparisons measure this tradeoff at scale; runtime components locate saved and remaining work; ablations isolate the two maintenance paths; parameter and seed studies examine when the savings persist. Finally, inverse-power experiments test the spectral weighting that motivated the broader objective family.

A. Datasets and protocol

a) Graphs and budget: The seven SNAP graphs [15] have the component sizes listed in Table III. Inputs are unweighted and symmetrized, with loops and duplicate edges removed, then stored in Cuthill–McKee order. Every vertex outside the common singleton seed is eligible. We use hard grounding, $\sigma = 0$, and $K = 20$ total anchors unless stated otherwise. The seed-stage Johnson–Lindenstrauss (JL) dimension is 128; selection uses $d = 64$, the empirical setting studied in Table VIII rather than the worst-case sufficient size of Theorem 2.

b) Compared methods: All baselines are our C++ reimplementations inside one engine. They share GROVE's graph storage, PCG solver, preconditioner, and thread settings, so same-machine time differences reflect the algorithms rather than languages or solver libraries; we located no released code for the forest methods [5]. ApproxGreedy is fresh-round all-node JL marginal estimation [1]. Its released code[1] accompanies the original publication and targets Julia 0.6 with the Laplacians.jl solver; we therefore reimplement it in the shared engine with JL dimension 64, the same as GROVE's d, and compare its marginals and selected sets with dense exact greedy on small graphs. ForestCFCM and SchurCFCM are Xia and Zhang's forest-sampling algorithms [5]. SchurCFCM was checked against dense exact values: fixed-set estimates converge over up to 10^5 forests, nine $k = 10$ selections have exact traces within 0.81% of exact greedy, and its auxiliary-root counts on DBLP, Amazon, and Skitter (131, 96, 965) equal those reported in [5]. Both forest methods use JL dimension 64 and input parameter $\varepsilon_{\text{forest}} = 0.2$. Their adaptive stopping rule did not terminate on these graphs, so each round draws at most 1400 forests, 100 per thread; a fourfold larger cap on DBLP leaves the comparison unchanged (Section VI-B). SchurCFCM uses the original degree-based auxiliary-root rule, producing between 8 auxiliary vertices on roadNet-PA and 965 on Skitter. Zhou is the Fast score [6] with warm-started inverse power iteration targeting $\lambda_{\min}$. All selectors receive the same initial anchor.

c) Execution and parameters: All methods run on one host with two Intel Xeon Gold 6330 CPUs at 2.0 GHz, 755 GB RAM, and Ubuntu 22.04. Each job is limited to 14 cores and 14 OpenMP threads. The C++17 engine is compiled by g++ 11.4.0 with `-O3 -march=native -std=c++17 -fopenmp`. Selection time includes factor/sketch construction, refinement, and updates; it excludes graph loading, the shared seed stage, and independent evaluation. The seed stage takes 3.747–62.708 seconds.

[1]https://github.com/lchc/CFCC-maximization

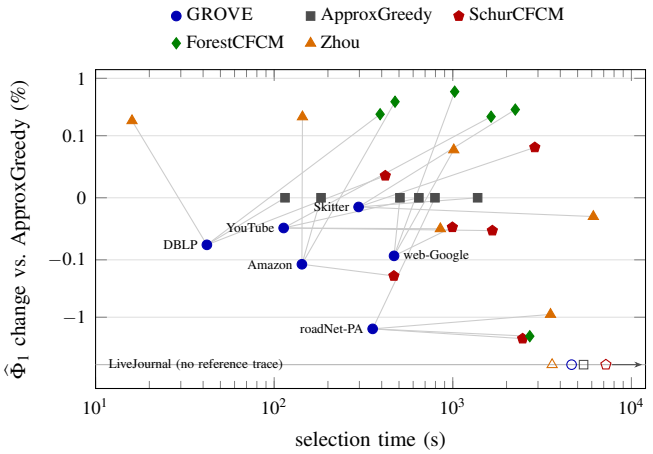


Fig. 2. Selection time against the relative change of estimated inverse trace with respect to ApproxGreedy on the same graph (negative is better); gray segments join GROVE to each baseline on the same graph. The vertical axis is symmetric-logarithmic: linear within ±0.01%, logarithmic beyond. On LiveJournal (bottom strip), hollow markers give the GROVE time and the censored baseline limits; the arrow marks censoring.

GROVE uses both update paths, selected-column reuse, and acceptance tolerance $\theta = 0$. Initialization and refresh batch widths are $B_{\text{init}} = 32$ and $B_r = 16$. The refresh cap b_r is 16 on roadNet-PA and 64 elsewhere. On Skitter no round exceeds 32 refreshes, so this cap is equivalent to unlimited refreshes. The seed preconditioner uses approximate Cholesky as described in Section IV; solve tolerances are $\varepsilon_{\text{init}} = 10^{-3}$, $\varepsilon_{\text{ref}} = 10^{-6}$, and $\varepsilon_{\text{upd}} = 10^{-9}$.

d) Evaluation: Let X be the final active grounded matrix. We estimate $\Phi_1 = \operatorname{tr} X^{-1}$ and $\operatorname{tr} X^{-2}$ with 128 common Rademacher probes [13] and compute $\lambda_{\min}(X)$ by Lanczos on X^{-1}. Evaluation solves use relative residual tolerance 10^{-7}. Lower estimated inverse trace $\widehat{\Phi}_1$ means higher CFCC. For a method pair, the paired gap is $\widehat{\Phi}_1(\text{other}) - \widehat{\Phi}_1(\text{GROVE})$, positive when GROVE is better. Its standard error (SE) measures evaluation noise, and $|\text{gap}|/\text{SE}$ is the standardized gap. We use $|\text{gap}|/\text{SE} > 2$ as a descriptive threshold for a detectable difference. Incomplete runs have a time limit but no final objective value.

B. Main comparison

Reusing state reduces selection time with small objective differences. On the six graphs with completed main comparisons, every baseline's inverse trace is within 1.6% of GROVE's, while times differ substantially (Table II, Fig. 2).

a) ApproxGreedy: The fresh-round comparison directly tests the cost of rebuilding estimates. GROVE is 1.1–5.7× faster on all six graphs: 5.7× on YouTube, 4.6× on Skitter, 2.7× on DBLP, and 1.1× on web-Google. Over four further selection seeds on four graphs (Table VI), the ratio of mean times is 1.4–3.8×. The YouTube main run (113 seconds) is faster than GROVE's multi-seed mean of 156 seconds, so its multi-seed ratio is 3.75× rather than 5.7×. GROVE's estimated

TABLE II

MAIN COMPARISON: HARD CFCC, $K = 20$, SELECTION $d = 64$, 14 THREADS, 128 COMMON EVALUATION PROBES. EACH METHOD HAS SELECTION TIME IN SECONDS AND ESTIMATED $\widehat{\Phi}_1 \approx \mathrm{tr} X^{-1}$ (LOWER IS BETTER). ALL METHODS START FROM THE SAME COMMON SEED; SCHURCFCM AND FORESTCFCM DRAW AT MOST 1400 FORESTS PER ROUND. FORESTCFCM WAS NOT RUN ON LIVEJOURNAL.

	GROVE		ApproxGreedy		SchurCFCM		ForestCFCM		Zhou	
Graph	Time	$\widehat{\Phi}_1$	Time	$\widehat{\Phi}_1$	Time	$\widehat{\Phi}_1$	Time	$\widehat{\Phi}_1$	Time	$\widehat{\Phi}_1$
DBLP	42	188268.0	115	188365	419	188390.6	392	188821.6	16	188716
Amazon	143	221167.8	183	221435	468	221003.3	475	222317.3	144	221917
YouTube	113	1091405.9	646	1091648	1666	1091368.5	1639	1094043.2	852	1091395.5
web-Google	469	479082.5	506	479486.8	995	479384.9	1025	482329.8	1013	479743.9
Skitter	297	676053.0	1378	676081.8	2879	676487.6	2239	678054.8	6124	676009.8
roadNet-PA	356	3.2230M	795	3.275M	2461	3.1992M	2699	3.2060M	3526	3.245M
LiveJournal	4626	1626554	>5400	–	>7200	–	–	–	>3600	–

TABLE III

MAIN DATASETS: LARGEST CONNECTED COMPONENTS AFTER PREPROCESSING.

Graph	n	m
com-DBLP	317,080	1,049,866
com-Amazon	334,863	925,872
com-YouTube	1,134,890	2,987,624
web-Google	855,802	4,291,352
as-Skitter	1,694,616	11,094,209
roadNet-PA	1,087,562	1,541,514
com-LiveJournal	3,997,962	34,681,189

inverse trace is lower on all six, with paired gaps exceeding two SE on five; Skitter's gap is 28.8 ± 22. The largest difference is on roadNet-PA: 3.2230×10^6 in 356 seconds versus 3.275×10^6 in 795 seconds.

b) Forest sampling: GROVE is 2.1–14.8× faster than the two forest methods under the 1400-forest cap. ForestCFCM has higher inverse traces on five graphs, but a 0.53% lower trace on roadNet-PA. SchurCFCM is lower on Amazon and roadNet-PA by 0.07% and 0.74%, at 3.3× and 6.9× the time; its YouTube difference is within two SE. It is higher on DBLP, web-Google, and Skitter. On DBLP, raising the forest budget to 5600 gives SchurCFCM a gap of -24.6 ± 11.5 in 905 seconds: a small 0.01% improvement slightly beyond two SE. ForestCFCM reaches $+235.7 \pm 24.4$ in 1591 seconds. A fourfold forest budget therefore changes neither the time ordering nor the scale of the quality differences.

c) Zhou: The spectral heuristic tests the effect of optimizing a different objective. Zhou is faster on DBLP (16 versus 42 seconds) and almost equally fast on Amazon (144 versus 143). GROVE is faster on the other four graphs, by 2.2–20.6×. Its inverse trace is lower on DBLP, Amazon, web-Google, and roadNet-PA. YouTube is within two SE, while Zhou is slightly better on Skitter (-43.1 ± 19.6, or 0.006% of the trace).

d) LiveJournal: On the largest input, GROVE finishes in 4626 seconds with estimated inverse trace 1626554. ApproxGreedy reaches its 5400-second limit during round 17, SchurCFCM its 7200-second limit after 14 of 19 rounds, and Zhou its 3600-second limit after three rounds. The limits of 90, 120, and 60 minutes were fixed per method before the runs, from pilot per-round costs on the shared server. The ApproxGreedy and SchurCFCM limits exceed GROVE's completion time, so both are slower on this input whatever the limit; Zhou's second round alone took 3140 seconds. The fixed-forest ForestCFCM batch covered the other six graphs, so its LiveJournal cell is unavailable. The scalar-cache ablation completes separately.

TABLE IV

MAIN-RUN RUNTIME COMPONENTS (SECONDS) AND SPMV COUNTS. SELECT IS TOTAL MINUS INIT AND UPDATE, NOT A SEPARATELY TIMED REFRESH KERNEL. G DENOTES GROVE AND AG DENOTES APPROXGREEDY; – IS AN INCOMPLETE BASELINE.

	GROVE time			SpMVs	
Graph	Init	Select	Update	G	AG
DBLP	5.7	25.7	10.7	15185	32672
Amazon	7.5	114.4	21.5	45766	55168
YouTube	20.1	69.0	23.7	13053	49856
web-Google	26.5	383.9	58.9	49618	48192
Skitter	41.6	144.8	110.7	18357	51872
roadNet-PA	36.3	259.1	60.4	23433	69536
LiveJournal	220.9	3856.0	549.1	43599	–

C. Runtime components and solve work

Table IV separates initialization, updates, and the remaining selection work. Initialization includes factor construction and sketch solves; updates include selected-column solves and sketch/scalar maintenance. The selection remainder includes refreshes, heap operations, and other overhead. Sparse matrix–vector products (SpMVs) are counted per right-hand side, not as batched PCG iterations.

The profiles expose distinct regimes. YouTube uses 13,053 SpMVs versus 49,856 for ApproxGreedy. On Skitter, update work already takes 110.7 seconds, or 37.3% of the total. On web-Google, GROVE uses 49,618 SpMVs versus 48,192 for ApproxGreedy: 640 candidate refreshes and three cap hits leave 383.9 seconds in the selection remainder (81.8% of runtime), consistent with the small 1.08× speedup. Solves of different widths and tolerances have different costs, so these counts describe workload rather than predict elapsed time.

On roadNet-PA every round reaches the cap at $K = 10, 20, 50$, giving 144, 304, and 784 refreshes. Per-round refresh work therefore persists as the budget grows. The speedup rises modestly from 2.08× to 2.54×; it is the paired quality gap, not the speed ratio, that narrows at $K = 50$ (Table VII).

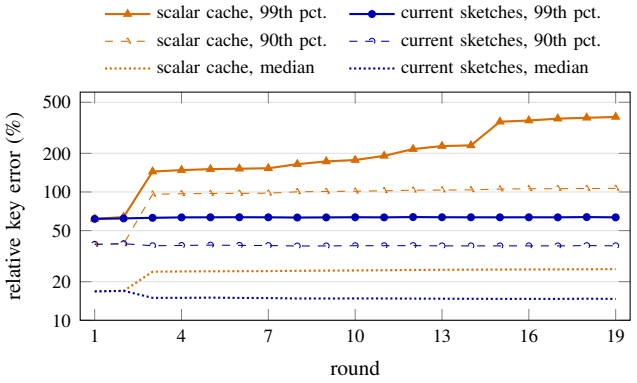


Fig. 3. HepPh, $d = 64$: relative error of the estimated key r_2/r_1 against exact moments before each round, over all active candidates (mean of five projection draws). The scalar cache keeps its initial absolute error; the current sketches used by GROVE for unrefreshed candidates keep a constant relative error.

D. Two complementary maintenance paths

Table V compares the two reuse paths on a common set of rows. Scalar caching starts from noisy sketch moments and retains their initial absolute errors under exact decrements. GROVE instead keeps current solution sketches for unrefreshed candidates and retains solved moments for refreshed candidates. The ablation measures selection time and final quality; Fig. 3 measures the per-candidate errors directly.

Scalar caching is competitive on most graphs. On the six inputs other than roadNet-PA its time is 0.95–1.15× that of GROVE, and its trace is equal within noise on DBLP, YouTube, and Skitter, 0.04% higher on Amazon, and lower on web-Google (paired gap -40.9 ± 3.3). It has no error control, however, and it fails on roadNet-PA: 881 seconds and 3.659×10^6, a trace 13.5% above GROVE's 3.2230×10^6 at 2.5× the time. The value of GROVE is that it avoids this failure without giving up speed on the other graphs.

Fig. 3 shows the mechanism on HepPh ($n = 11{,}204$), where dense inverses give exact moments. Both estimators start from the same $d = 64$ sketches and follow the exact-greedy trajectory from the common seed. The current sketches keep the relative error of the key r_2/r_1 flat: its 90th and 99th percentiles stay at 38% and 63% through all 19 rounds. The scalar cache preserves its initial absolute error exactly while moments near the new anchors shrink, so the same percentiles rise to 107% and 384%. Refreshed GROVE candidates carry exact moments.

Sketch-only removes retention. It takes 424 seconds on YouTube and 570 on Skitter, versus 113 and 297 for GROVE, with nearly identical trace estimates. On Skitter, 12 of 19 hybrid rounds need no refresh; only 128 candidates are ever refreshed. Independent sketch-only results cover these two graphs and roadNet-PA; missing cells are shown explicitly. LiveJournal's completed scalar-cache set was not evaluated. The cap-64 roadNet-PA sketch-only result is discussed below.

E. Sensitivity to budget, dimension, and refresh cap

We vary K, d, and b_r to test the balance between reuse, estimate quality, and refinement work. Tables VII and VIII vary one parameter at a time on DBLP, YouTube, and roadNet-PA; refresh totals count distinct candidates.

a) Budget: More rounds increase the opportunity for reuse. From $K = 10$ to $K = 50$, the speedup over ApproxGreedy rises from 2.0× to 4.3× on DBLP and 4.2× to 6.5× on YouTube; it remains 2.1–2.5× on roadNet-PA. GROVE has lower estimated inverse trace at every tested budget. The gaps exceed four SE except at $K = 10$ on DBLP and YouTube, where they are about two SE.

b) Sketch dimension: A narrower solution sketch can increase refinement cost. Raising d from 16 to 64 reduces YouTube's refreshes from 784 to 144 and time from 407 to 113 seconds; DBLP shows the same pattern. At $d = 128$, trace differences from $d = 64$ are within two SE on all three graphs. On roadNet-PA the cap binds throughout the sweep, and time instead rises mildly with dimension, from 315 to 365 seconds.

c) Refresh cap: The cap trades refinement against quality. On roadNet-PA, sketch-only caps 16 and 64 take 278 and 1069 seconds, reaching 3.2366×10^6 and 3.2098×10^6. The hybrid with cap 16 lies between them in quality, at 3.2230×10^6 in 356 seconds. On Skitter the cap is inactive. These cases separate bounded refinement from the benefits of the two update paths.

F. Variability and paired differences

Selection randomness and evaluation noise measure different sources of variation. Table VI uses four further selection seeds (1001–1004) with a fixed initial anchor. Population standard deviations summarize these runs; speedups are ratios of mean times. GROVE remains faster on all four graphs, by 1.39–3.75×, and has lower estimated inverse trace in all 16 graph–seed pairs. Fifteen paired gaps exceed three SE; the remaining YouTube pair has gap/SE $= 0.7$.

Table X evaluates the main-run sets with common probes to reduce noise in the differences. For example, the DBLP comparison with ApproxGreedy has paired SE 16.7, versus 82 for a single trace estimate. These paired errors support the main quality comparisons; the repeated-selection standard deviations describe run-to-run variation separately.

G. The inverse power as a tunable knob

This experiment tests spectral weighting independently of the large-scale $p = 1$ sketch implementation. On five medium graphs, exact-marginal lazy selection uses $p = 1, 2, 3, 4, 8$, with moments computed by numerical solves. Final traces use unit-vector solves at relative residual tolerance 10^{-9}, and smallest eigenvalues use Lanczos (Table IX, Fig. 4).

From $p = 1$ to $p = 4$, $\lambda_{\min}$ increases by 1.51–2.33×, while inverse trace rises only 0.05–0.60%. Among the tested powers, $p = 1$ gives the lowest trace on each graph. The spectral response is not strictly monotone: CAIDA has a 0.4% dip from $p = 2$ to $p = 3$. At $p = 4$, the smallest eigenvalue is within 0.1% of Zhou's on AstroPh and Enron, and 4.5–9.5% lower on the other graphs. At $p = 8$, it exceeds Zhou's on four graphs and is 0.05% lower on Enron. These sets expose the cost of emphasizing slower modes rather than treating either objective as interchangeable with CFCC.

TABLE V

UPDATE-PATH ABLATIONS: TIME (S) / ESTIMATED INVERSE TRACE. NA MEANS AN INDEPENDENT SKETCH-ONLY RUN IS UNAVAILABLE; – MEANS A COMPLETED SET WAS NOT EVALUATED. SKETCH-ONLY USES CAP 16 ON ROADNET-PA; ITS YOUTUBE AND SKITTER CAPS ARE NONBINDING. GROVE USES THE MAIN COMPARISON SETTINGS.

Graph	Scalar cache	Sketch-only	GROVE
DBLP	40 / 188268	NA	42 / 188268.0
Amazon	136 / 221259	NA	143 / 221167.8
YouTube	130 / 1091406	424 / 1091405.9	113 / 1091405.9
web-Google	490 / 479041.6	NA	469 / 479082.5
Skitter	288 / 676053.6	570 / 676052.1	297 / 676053.0
roadNet-PA	881 / 3.659M	278 / 3.2366M	356 / 3.2230M
LiveJournal	5206 / –	NA	4626 / 1626554

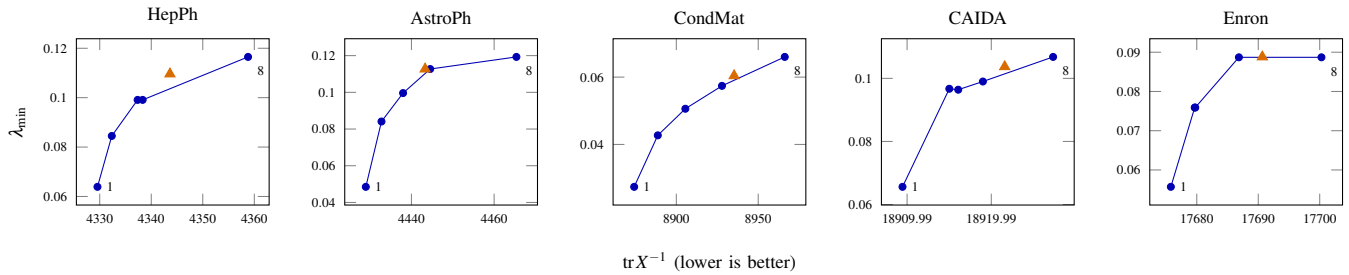


Fig. 4. CFCC/spectral tradeoff on the medium graphs of Table IX. Circles join the sets selected at $p = 1,2,3,4,8$ (labels mark $p = 1$ and $p = 8$); the triangle is Zhou's single set.

TABLE VI

FOUR FURTHER SEEDS OF THE SELECTION RANDOMNESS, $K = 20$, $d = 64$, FIXED COMMON SEED VERTEX: MEAN ± STANDARD DEVIATION OF SELECTION TIME (S) AND OF THE PAIRED GAP $(\widehat{\Phi}_1(\text{APPROXGREEDY}) - \widehat{\Phi}_1(\text{GROVE}))/\widehat{\Phi}_1$; THE LAST COLUMN GIVES THE RANGE OF PER-SEED PAIRED GAP/SE RATIOS.

Graph	GROVE	ApproxGreedy	Speedup	Gap (%)	Gap/SE
DBLP	37.6 ± 3.4	123.9 ± 2.3	3.29×	0.035 ± 0.009	3.4–5.5
Amazon	136.2 ± 9.2	189.6 ± 8.8	1.39×	0.219 ± 0.047	5.4–10.0
YouTube	156.0 ± 19.6	585.5 ± 41.8	3.75×	0.019 ± 0.010	0.7–8.1
roadNet-PA	343.5 ± 7.9	793.6 ± 18.8	2.31×	1.191 ± 0.078	5.8–7.5

TABLE VII

TOTAL BUDGET K ($d = 64$): SELECTION TIME (S) AND PAIRED DIFFERENCE $\widehat{\Phi}_1(\text{APPROXGREEDY}) - \widehat{\Phi}_1(\text{GROVE}) \pm$ SE; POSITIVE VALUES FAVOR GROVE.

Graph	K	GROVE	ApproxGreedy	Speedup	Difference
DBLP	10	24.3	49.2	2.0×	$+23.8 \pm 12.2$
	20	42.1	115	2.7×	$+97.2 \pm 16.7$
	50	92.7	402.6	4.3×	$+160.0 \pm 17.3$
YouTube	10	56.1	237.4	4.2×	$+65.5 \pm 32.7$
	20	112.9	646	5.7×	$+241.7 \pm 44.3$
	50	346.7	2260.3	6.5×	$+349.8 \pm 43.5$
roadNet-PA	10	160.1	333.6	2.1×	$+48575 \pm 10477$
	20	355.8	795	2.2×	$+51513 \pm 5865$
	50	1020.4	2590.4	2.5×	$+18562 \pm 3366$

VII. RELATED WORK

a) Group centrality and grounded spectra: Li et al. [1] formulate CFCC, establish optimization hardness and reciprocal-objective supermodularity, and develop exact and randomized greedy algorithms, including ApproxGreedy. Current-flow closeness of single vertices goes back to Brandes and Fleischer [17], and group centralities to Everett and Borgatti [18]. Greedy and local-search maximization has been engineered for shortest-path group closeness and group harmonic centrality [19], [20], group betweenness by sampling [21], and group forest closeness [22]. Group Markov centrality considers related hitting-time group objectives [7]. Zhou et al. [6] instead maximize the smallest grounded eigenvalue, establishing monotonicity and failure of submodularity, and propose the Fast sensitivity score. Aggregate resistance and the slowest spectral mode capture different aspects of a selected group. Our CM formulation places CFCC in a trace family with tunable inverse-power weighting and an explicit connection to the spectral endpoint.

b) Leader selection: In networked control, choosing k noise-free leaders to minimize the steady-state variance of consensus followers is the same grounded inverse-trace problem [23]. Clark et al. [24] prove its supermodularity

TABLE VIII

SELECTION SKETCH DIMENSION d ($K = 20$): SELECTION TIME (S), DISTINCT REFRESHED CANDIDATES, AND PAIRED DIFFERENCE $\widehat{\Phi}_1(d) - \widehat{\Phi}_1(64) \pm$ SE.

	$d = 16$	$d = 32$	$d = 64$	$d = 128$
DBLP time	56.9	46.3	42.1	38.7
refreshes	512	368	256	176
difference	$+50.7 \pm 8.8$	$+4.3 \pm 10.4$	–	-4.7 ± 9.2
YouTube time	406.7	240.8	112.9	115.6
refreshes	784	448	144	96
difference	$+254.9 \pm 45.9$	$+160.1 \pm 33.3$	–	$+21.3 \pm 39.5$
roadNet-PA time	314.8	324.2	355.8	365.1
refreshes	304	304	304	304
difference	$+58297 \pm 6725$	$+22505 \pm 5047$	–	-9086 ± 4940

TABLE IX

EXACT-MARGINAL LAZY SELECTION ON MEDIUM GRAPHS, $K = 20$, COMMON SEED, $\sigma = 0$. EACH ENTRY IS $\lambda_{\min} / \operatorname{tr} X^{-1}$, EVALUATED EXACTLY: THE TRACE SUMS UNIT-VECTOR SOLVES AT RELATIVE RESIDUAL 10^{-9}, AND $\lambda_{\min}$ IS COMPUTED BY LANCZOS. ZHOU USES OUR FAST REIMPLEMENTATION.

n/m	HepPh 11,204 / 117,619	AstroPh 17,903 / 196,972	CondMat 21,363 / 91,286	CAIDA 26,475 / 53,381	Enron 33,696 / 180,811
$p = 1$	.0639 / 4329.56	.0485 / 4428.97	.0274 / 8874.27	.0656 / 18909.45	.0557 / 17675.78
$p = 2$	.0845 / 4332.34	.0841 / 4432.76	.0427 / 8888.70	.0967 / 18915.01	.0759 / 17679.66
$p = 3$	.0991 / 4337.35	.0996 / 4437.98	.0506 / 8905.52	.0964 / 18916.07	.0759 / 17679.75
$p = 4$	.0991 / 4338.33	.1127 / 4444.54	.0574 / 8927.88	.0990 / 18919.00	.0887 / 17686.85
$p = 8$	.1165 / 4358.79	.1193 / 4465.39	.0660 / 8966.20	.1068 / 18927.34	.0887 / 17700.28
Zhou	.1096 / 4343.63	.1127 / 4443.31	.0604 / 8935.30	.1037 / 18921.59	.0888 / 17690.66

TABLE X

PAIRED-PROBE DIFFERENCES $\widehat{\Phi}_1(\cdot) - \widehat{\Phi}_1(\text{GROVE})$ ON THE FINAL SETS OF TABLE II (128 COMMON PROBES), WITH STANDARD ERRORS OF THE DIFFERENCE. POSITIVE VALUES FAVOR GROVE; ROADNET-PA ENTRIES ARE IN THOUSANDS.

Graph	ApproxGreedy	ForestCFCM	SchurCFCM	Zhou
DBLP	$+97.2 \pm 16.7$	$+553.5 \pm 30.3$	$+122.5 \pm 14.2$	$+448 \pm 33$
Amazon	$+266.9 \pm 47.6$	$+1149.6 \pm 74.5$	-164.5 ± 47.6	$+749 \pm 85$
YouTube	$+241.7 \pm 44.3$	$+2637.2 \pm 81.4$	-37.4 ± 44.8	-10.5 ± 39
web-Google	$+404 \pm 79$	$+3247.3 \pm 143.2$	$+302.4 \pm 83.5$	$+661 \pm 98$
Skitter	$+28.8 \pm 22$	$+2001.9 \pm 66.9$	$+434.6 \pm 29.5$	-43.1 ± 19.6
roadNet-PA	$+51.5 \pm 5.9$	-17.0 ± 5.8	-23.8 ± 5.8	$+22.2 \pm 5.6$

and analyze greedy selection; Lin et al. [25] give convex relaxations and bounds. These methods use dense inverses and target networks of hundreds to thousands of nodes. The smallest eigenvalue of the grounded Laplacian, which governs convergence to the leaders, is analyzed by Pirani and Sundaram [26].

c) Forest sampling for CFCC: Xia and Zhang [5] estimate CFCC marginals using spanning forests rooted at the selected group and generated by Wilson's algorithm [4]. ForestCFCM estimates marginals directly; SchurCFCM introduces auxiliary high-degree roots and applies Schur-complement corrections. Both draw new forests after each selection and use graph-dependent bounds for adaptive sampling. GROVE instead maintains solution sketches across grounding updates and retains refreshed moments, reducing repeated all-candidate estimation.

d) Spectral design and matrix functions: FastFrac [8] optimizes positive fractional traces under edge addition; POLAR [9] studies negative fractional objectives through submodularity-ratio analysis. Edge modification changes walk transition rates, whereas grounding changes survival by killing or removing active coordinates. Matrix-function representations and quadrature [14] evaluate spectral objectives through resolvents. Our integer formulation uses inverse moments for finite grounding marginals and their incremental updates.

e) Randomized and incremental numerical methods: JL projections reduce inverse-diagonal estimation to a small number of solves, as in effective-resistance sketching [2] and ApproxGreedy [1]. Sherman–Morrison identities [16] give rank-one inverse updates, while classical lazy greedy [11] uses stale submodular upper bounds. GROVE combines current sketches, retained moments, and capped refresh; its analysis handles reuse of the same projections across adaptive rounds. Nearly-linear solvers and sparse elimination [3], [12] provide the numerical primitives.

VIII. CONCLUSION

GROVE reduces repeated CFCC estimation by updating solution sketches and retaining refreshed moments. Its seeded-gain analysis makes adaptive sampling and numerical conditions explicit. The measurements show both substantial solve savings and limits from maintenance or refresh work, with input-dependent quality tradeoffs. The CM and inverse-power extension supplies a separate choice of spectral objectives.

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APPENDIX A
COMPLETE CM GROUNDING PROOF

This appendix supplies the detailed statements underlying Theorem 4. We use A_S^β from (1), the admissible seed lattice from Section II, and $a = \mu(\{0\}) = f(\infty)$. The diminishing-drop inequality is

$$\Phi(S) - \Phi(S+v) \geq \Phi(T) - \Phi(T+v), \quad S \subseteq T,\ v \notin T. \quad (12)$$

Lemma A1 (One coupling for all grounding sets)**.** *There is a coupling, independent of the selected set, with killing times T_S such that $\mathsf{T}_{S\cup T} = \min(\mathsf{T}_S, \mathsf{T}_T)$. For soft grounding,*

$$(e^{-tA_S^\beta})_{ij} = \mathbb{P}_i(\mathsf{X}_t = j,\ t < \mathsf{T}_S), \quad t \geq 0.$$

For hard grounding the same identity holds after embedding $e^{-tA_S^\infty}$ into an $n \times n$ matrix by zeros in rows and columns in S, including at $t = 0$.

Proof. Let X be the continuous-time walk on V with jump rate w_{ij} from i to j, holding rate $d_i = \sum_j w_{ij}$, and generator $-L$. It exists for all time since the state space and all rates are finite; an isolated state has infinite holding time. Independently sample $E_0 \sim \mathrm{Exp}(\sigma)$ (take $E_0 = \infty$ if $\sigma = 0$), and $E_v \sim \mathrm{Exp}(\beta)$ independently for every v. Define occupation times $\ell_v(t) = \int_0^t \mathbf{1}_{\{\mathsf{X}_u = v\}} du$ and

$$\kappa_v = \inf\{t : \ell_v(t) \geq E_v\}, \qquad \mathsf{T}_S = E_0 \wedge \min_{v\in S} \kappa_v.$$

These clocks run in local occupation time. In particular an unvisited vertex cannot kill the walk. The displayed minimum identity holds pathwise, simultaneously for all sets. Conditioning on the walk gives

$$\mathbb{P}(t < \mathsf{T}_S \mid \mathsf{X}) = e^{-\sigma t} \exp\left(-\beta \sum_{v\in S} \ell_v(t)\right).$$

The exponential memoryless property implies that while at i the additional killing rate is $\sigma + \beta \mathbf{1}_S(i)$. The transient transition matrix $K_S(t)$ therefore has initial value I and generator $Q_{ij} = w_{ij}$ for $i \neq j$, $Q_{ii} = -d_i - \sigma - \beta \mathbf{1}_S(i)$. The Markov property and the first-step expansion give $K_S'(t) = K_S(t)Q$; uniqueness of this finite linear ODE yields $K_S(t) = e^{-tA_S^\beta}$.

For hard grounding replace κ_v by the hitting time $H_v = \inf\{t \geq 0 : \mathsf{X}_t = v\}$ and keep the same E_0 and base walk. On $U = V \setminus S$ the transient rates are w_{ij}, $i, j \in U$, $i \neq j$, and diagonal $-d_i - \sigma$; jumps into S cause absorption. Thus the transient generator is $-M[U,U]$. A start in S has $\mathsf{T}_S = 0$ and contributes zero even at $t = 0$. A surviving walk cannot end in S. These facts give precisely the zero embedding and prove the identity for every i, j. All the constructions use one base walk and the same independent clocks for every set. □

Theorem A2 (CM grounding traces)**.** *For every CM f and $\beta \in (0,\infty]$, Φ_f^β is nonincreasing and supermodular on the admissible lattice. For each admissible S and $v \notin S$, the drop is strictly positive in the soft case exactly when f is nonconstant, and in the hard case exactly when $f \not\equiv 0$. The soft-to-hard limit is*

$$\lim_{\beta\to\infty} \Phi_f^\beta(S) = \Phi_f^\infty(S) + |S|a. \quad (13)$$

In particular, the two objectives agree in the limit when $f(\infty) = 0$.

Proof. Fix a starting vertex i, time t, and one coupled outcome. Define

$$Z_S = \mathbf{1}_{\{\mathsf{X}_t = i\}} \mathbf{1}_{\{t < \mathsf{T}_S\}}.$$

For $S \subseteq T$, $Z_S \geq Z_T$. For the added vertex clock κ_v (or hitting time in the hard case),

$$Z_S - Z_{S\cup\{v\}} = \mathbf{1}_{\{\mathsf{X}_t = i\}} \mathbf{1}_{\{t < \mathsf{T}_S\}} \mathbf{1}_{\{\kappa_v \leq t\}}.$$

Replacing S by T only decreases this product. Taking expectations and summing over all $i \in V$ proves monotonicity and (12) for $h_S(t) = \mathrm{tr}\, e^{-tA_S^\beta}$, using the zero embedding in the hard case. This argument includes $t = 0$: then the soft heat trace is n and the hard heat trace is $n - |S|$.

Diagonalizing each positive definite matrix and using Tonelli's theorem for the nonnegative scalar integrands gives

$$\Phi_f^\beta(S) = \int_{[0,\infty)} h_S(t)\, \mu(dt) = \sum_i \int_{[0,\infty)} \mathbb{E}_i Z_S\, \mu(dt).$$

The measure is sigma-finite, since $\mu([0,j]) \leq e^j f(1)$. Every integral is finite by the finite spectral sum. Differences can therefore be taken, or equivalently Tonelli can be applied directly to the nonnegative drops. The pointwise drop inequalities integrate to the claimed inequalities without subtracting infinities.

For strictness, start at $v \notin S$. The event of no jump before t and survival of base killing has probability $e^{-(d_v+\sigma)t}$. For soft grounding it contributes $e^{-(d_v+\sigma)t}(1 - e^{-\beta t}) > 0$ to the heat-trace drop for every $t > 0$. Thus any mass of μ on $(0,\infty)$ gives a strict drop; if there is no such mass $f = a$ and the soft objective is na. The representation shows that f is constant exactly in this latter case. For hard grounding the no-jump event contributes $e^{-(d_v+\sigma)t} > 0$, and the drop at $t = 0$ is one. Hence every nonzero μ gives a strict drop; $\mu = 0$ gives the zero objective.

For the limit use common unit exponential variables and set $E_v^\beta = E_v^1/\beta$. For fixed $t > 0$, the conditional survival factor converges to the indicator that no selected vertex has positive occupation time before t. Apart from a null event of a jump exactly at t, this is hard survival; a start in S also has positive occupation time. Bounded convergence gives $h_S^\beta(t) \to h_S^\infty(t)$ for $t > 0$. At $t = 0$ the limit remains n, whereas the hard trace is $n - |S|$. Choose a fixed $\beta_0 > 0$. For $\beta \geq \beta_0$, $0 \leq h_S^\beta(t) \leq h_S^{\beta_0}(t)$, an integrable bound by admissibility. Dominated convergence proves (13), including $S = V$. □

Corollary A3 (Powers, heat, and current-flow closeness)**.** *The theorem applies to $f(x) = x^{-s}$ for every $s > 0$, to $f(x) = e^{-\tau x}$ for $\tau \geq 0$, and to $(x+c)^{-1}$ for $c \geq 0$. For a connected graph, $\sigma = 0$, and $\varnothing \neq S \subsetneq V$, the group current-flow closeness is $\mathrm{CFCC}(S) = n/\,\mathrm{tr}\, L_{-S}^{-1}$. Thus the reciprocal-CFCC supermodularity of Li et al. [1] is the hard-grounding $s = 1$ case.*

Proof. The positive representations are, respectively,

$$x^{-s} = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-tx} dt,$$
$$e^{-\tau x} = \int e^{-tx} \delta_\tau(dt),$$
$$(x+c)^{-1} = \int_0^\infty e^{-tx} e^{-ct} dt.$$

The first follows by $u = tx$ in the defining Gamma integral, and the third is an elementary exponential integral. Apply Theorem A2. For the electrical identity, ground S and inject a unit current at $u \notin S$. Kirchhoff's equations on $U = V \setminus S$ are $L[U,U]p = e_u$. Thus the source voltage, equal to its effective resistance to S, is $(L[U,U]^{-1})_{uu}$. Summing these voltages and using the definition of group current-flow closeness gives the formula. Multiplication by $1/n$ preserves supermodularity of the reciprocal objective. □

Proposition A4 (Edge addition has a different structure)**.** *Even for $f(x) = x^{-1}$, edge addition can violate supermodularity on a connected five-vertex graph.*

Proof. Take the unit-weight path $2-1-0-4-3$, $\sigma = 1/10$, and candidate unit edges $e = \{0,2\}$, $g = \{0,3\}$. Write $H(B) = \operatorname{tr}(L + \sum_{b \in B} b_b b_b^\top + I/10)^{-1}$, where $b_{\{u,v\}} = e_u - e_v$. Exact rational inversion gives

$$H(\varnothing) = \frac{966250}{72181}, \qquad H(\{e,g\}) = \frac{204350}{17391},$$
$$H(\{e\}) = H(\{g\}) = \frac{2486050}{197501}.$$

Rational Gaussian elimination gives these values; the inverse identities are verified by direct matrix multiplication. Their mixed difference is exactly

$$H(\{e,g\}) - H(\{e\}) - H(\{g\}) + H(\varnothing) = -\frac{9866618000}{257984349711} < 0.$$

For grounding the off-diagonal transition rates remain fixed and only killing changes. Adding an edge instead changes two transition rates, hence changes the base path and its return event. Additional jumps cannot be represented by deleting trajectories with independent vertex killing clocks. Algebraically the edge direction $e_u - e_v$ has both signs, unlike the grounding direction e_v; for $R = A^{-1}$, direct differentiation gives $\partial_x \partial_y \operatorname{tr}(A + xbb^\top + ycc^\top)^{-1}|_0 = 2(b^\top Rc)(b^\top R^2 c)$, which has no fixed sign. Indeed $\partial_x \operatorname{tr} R = -b^\top R^2 b$ and $\partial_y R = -Rcc^\top R$ give the displayed derivative by the product rule. This identifies exactly the unavailable step in the coupling proof for edge addition. □

Appendix B
Hardness Proofs

The first four results prove the inverse-power and heat parts of the selection-hardness claims discussed in Section V-A, together with the finite-penalty variants. Here the reduction's k denotes its specified total cardinality; a singleton-seed call uses $k-1$ additional choices. The second part proves the stronger exact-budget spectral barrier.

Lemma A5 (Cubic vertex-cover gap)**.** *Let G be a connected unit-weight 3-regular graph, let $\varnothing \neq S \subsetneq V$, put $r = n - |S|$, and let e_U count the edges induced by $U = V \setminus S$. For every $s > 0$ and $\sigma \geq 0$,*

$$\operatorname{tr}(L_{-S} + \sigma I)^{-s} \geq r(3+\sigma)^{-s} + \kappa^{\mathrm{curv}}_{s,\sigma} e_U,$$
$$\kappa^{\mathrm{curv}}_{s,\sigma} = \frac{s(s+1)}{(6+\sigma)^{s+2}} > 0. \tag{14}$$

Equality with the baseline $r(3+\sigma)^{-s}$ holds exactly when S is a vertex cover. Also $\lambda_{\min}(L_{-S} + \sigma I) \geq n^{-2}$.

Proof. Write $C = (3+\sigma)I - \mathsf{W}_U$, where W_U is the induced adjacency matrix. The eigenvalues of C lie in $(0, 6+\sigma]$: positive definiteness follows from the nonempty Dirichlet boundary, and $\|\mathsf{W}_U\|_2 \leq 3$. For $f(x) = x^{-s}$, throughout this interval $f''(x) \geq \kappa^{\mathrm{curv}}_{s,\sigma}$. Taylor's formula with integral remainder therefore gives

$$f(x) \geq f(b) + f'(b)(x-b) + \tfrac{1}{2}\kappa^{\mathrm{curv}}_{s,\sigma}(x-b)^2, \qquad b = 3+\sigma.$$

Sum this inequality over the eigenvalues. The linear terms vanish because $\operatorname{tr} C = rb$, and the sum of squared deviations is $\operatorname{tr} \mathsf{W}_U^2 = 2e_U$, proving (14). If $e_U = 0$, $C = bI$ gives equality; if $e_U \geq 1$ the excess is at least $\kappa^{\mathrm{curv}}_{s,\sigma}$. The equivalence with a vertex cover follows from the definition.

For the lower eigenvalue bound, extend $x \in \mathbb{R}^U$ by zeros on S. For each $u \in U$ choose a simple path from u to S, of length at most $n-1$. Cauchy–Schwarz bounds $|x_u|^2$ by n times the sum of edge differences squared along that path, hence by $nx^\top Lx$. Summing over at most n vertices yields $\|x\|^2 \leq n^2 x^\top Lx \leq n^2 x^\top (L + \sigma I)x$. The Rayleigh quotient proves the asserted eigenvalue bound. □

Lemma A6 (Uniform finite-penalty approximation)**.** *Under the hypotheses of Lemma A5, for $0 < s < 1$ and $B > 0$,*

$$0 \leq \operatorname{tr}(L + \sigma I + B \operatorname{diag}(\mathbf{1}_S))^{-s} - \operatorname{tr}(L_{-S} + \sigma I)^{-s}$$
$$\leq n \left(\frac{1 + 9n^4}{B} \right)^s. \tag{15}$$

Proof. Order the vertices as U, S and write

$$L + \sigma I = \begin{pmatrix} C & D \\ D^\top & E \end{pmatrix}, \qquad W = \begin{pmatrix} -C^{-1}D \\ I \end{pmatrix},$$
$$F = E + BI - D^\top C^{-1} D.$$

Because $L + \sigma I \succeq 0$ and $C \succ 0$, its Schur complement is positive semidefinite. For completeness, its quadratic form at y is the quadratic form of $L + \sigma I$ at $(-C^{-1}Dy, y)$. Consequently $F \succeq BI$. Block multiplication verifies

$$(L + \sigma I + B \operatorname{diag}(\mathbf{1}_S))^{-1} = J + WF^{-1}W^\top, \qquad J = \begin{pmatrix} C^{-1} & 0 \\ 0 & 0 \end{pmatrix}.$$

Thus $0 \preceq A^{-1} - J \preceq \varepsilon I$, where

$$\varepsilon = \frac{1 + \|C^{-1}D\|_2^2}{B} \leq \frac{1 + 9n^4}{B}.$$

Here $\|C^{-1}\|_2 \leq n^2$ by Lemma A5, and $\|D\|_2 \leq 3$ since it is a submatrix of the cubic adjacency matrix. The ordered

eigenvalues satisfy $\lambda_j(J) \le \lambda_j(A^{-1}) \le \lambda_j(J)+\varepsilon$ by the min–max principle. For $x \ge 0$ and $0 < s < 1$, $0 \le (x+\varepsilon)^s - x^s \le \varepsilon^s$: the difference is decreasing in $x > 0$, since $s-1 < 0$, and its maximum is attained at $x = 0$. Apply this scalar inequality to the matched eigenvalues and sum all n terms, including the $|S|$ zeros of J. Since $\operatorname{tr} J^s = \operatorname{tr} C^{-s}$, this proves (15). □

Theorem A7 (Hardness for fixed fractional exponent)**.** *For each fixed $s \in (0,1)$, finding an exact minimizing set of total cardinality k is NP-hard in each of the following regimes:*

1) *hard grounding on connected unit-weight cubic graphs, with any fixed $\sigma \ge 0$;*
2) *soft grounding on the same graphs, with any fixed $\sigma \ge 0$ and an input integer penalty β of polynomial magnitude;*
3) *soft grounding with $\sigma = 0$ and any fixed positive rational penalty β_0, on connected rationally weighted cubic graphs.*

For $\sigma = 0$, the specified-singleton-seed formulation is also NP-hard under polynomial-time Turing reductions.

Proof. Use vertex cover on connected 3-regular graphs, the NP-complete source problem used in Section 4 of Li et al. [1]. The nontrivial instances have $1 \le k < n$. A cover of size at most k exists exactly when one of size k exists, since a cover can be enlarged. Lemma A5 proves that whenever a size-k cover exists, every hard-objective minimizer is a cover. An optimal-set oracle therefore decides vertex cover by returning a set whose edge coverage can be checked in polynomial time. This proves item 1 without needing an exact representation of a real objective threshold.

For item 2, write $\delta = \kappa^{\mathrm{curv}}_{s,\sigma}$. Fix integers $p > 1/s$ and $C > 0$ such that $C^{-s} \le \delta/8$, and choose

$$B = C(1+9n^4)n^p.$$

These integers depend only on the fixed parameters s, σ. The chosen B has polynomial magnitude and $O(\log n)$ bits. By Lemma A6 the error for every nonempty proper set is at most $nC^{-s}n^{-ps} \le \delta/8$. Put $b = (n-k)(3+\sigma)^{-s}$. A cover has soft value in $[b, b+\delta/8]$, whereas a noncover has soft value at least $b+\delta$. Every soft minimizer is therefore a cover whenever one exists. Again check the returned set's edge coverage rather than comparing real-valued oracle outputs.

For item 3 use the same construction at $\sigma = 0$, and replace every unit edge weight by $c = \beta_0/B$. The new Laplacian is cL, and exactly

$$cL + \beta_0 \operatorname{diag}(\mathbf{1}_S) = c(L + B\operatorname{diag}(\mathbf{1}_S)),$$
$$\operatorname{tr}(cL + \beta_0 \operatorname{diag}(\mathbf{1}_S))^{-s} = c^{-s} \operatorname{tr}(L + B\operatorname{diag}(\mathbf{1}_S))^{-s}.$$

The positive common factor preserves all minimizers. The rational weights have $O(\log n)$ bits, and the support remains connected and cubic. This is a polynomial reduction with fixed finite β_0.

Finally suppose the oracle requires a specified nonempty seed at $\sigma = 0$. Make one call for each singleton seed $\{v\}$ with the same total cardinality k (or $k-1$ additional choices). If a size-k cover exists, one such seed belongs to it, and the separation arguments above ensure that the corresponding oracle returns a cover. If no cover exists, no returned set passes the edge coverage test. These n calls are a polynomial-time Turing reduction. □

Corollary A8 (Strictly convex hard-grounding objectives)**.** *For a fixed C^2 function f with $f''(x) \ge c_f > 0$ on $(0, 6+\sigma]$, the cubic-graph objective obeys*

$$\operatorname{tr} f(L_{-S} + \sigma I) \ge (n-|S|) f(3+\sigma) + c_f e_U.$$

Equality with the baseline characterizes vertex covers. Consequently, hard-grounding optimal-set selection is NP-hard for every fixed $s > 0$ with $f(x) = x^{-s}$, and for every fixed $\tau > 0$ with $f(x) = e^{-\tau x}$, at any fixed $\sigma \ge 0$. The singleton-seed qualification in Theorem A7 applies.

Proof. Repeat the Taylor inequality in Lemma A5 with c_f: the linear terms sum to zero and the quadratic terms sum to $c_f \operatorname{tr} \mathsf{W}_U^2/2 = c_f e_U$. A cover has $\mathsf{W}_U = 0$ and achieves the baseline, whereas a noncover exceeds it by at least c_f. Thus an optimal-set oracle decides the connected cubic vertex-cover problem exactly as in Theorem A7. Powers use $c_f = s(s+1)(6+\sigma)^{-s-2}$ for every $s > 0$; heat uses $c_f = \tau^2 e^{-\tau(6+\sigma)} > 0$. Checking returned sets avoids real threshold encoding. The same singleton-seed calls prove the seeded assertion. At $\tau = 0$ the fixed-cardinality heat objective is constant, so the heat statement requires $\tau > 0$. □

The following construction leaves the anchor budget unchanged and allows every new internal vertex to be selected. For an integer $\ell \ge 2$, define

$$a_\ell = 2 - 2\cos\frac{\pi}{2\ell-1}, \qquad b_\ell = 2 - 2\cos\theta_\ell, \tag{16}$$

where θ_ℓ is the unique root in $(0, \pi/(2\ell-1))$ of

$$3\cos((\ell - \tfrac{1}{2})\theta) = \cos(\theta/2). \tag{17}$$

Lemma A9 (Parallel-path YES/NO bounds)**.** *Let H be a connected simple cubic graph. Form $G_{m_{\mathrm{path}},\ell}$ by replacing every edge of H by m_{path} internally vertex-disjoint paths of length ℓ, with unit edge weights. The original vertices remain as the paths' endpoints. Then:*

1) *If S is an original vertex cover with $|S| < |V(H)|$, then*

$$\lambda_{\min}(L(G_{m_{\mathrm{path}},\ell})_{-S}) \ge \frac{a_\ell}{1 + 2(1+a_\ell)/(3m_{\mathrm{path}})}. \tag{18}$$

2) *Let T be any set of k vertices of $G_{m_{\mathrm{path}},\ell}$, including arbitrary internal vertices. If $T \cap V(H)$ is not a cover and $m_{\mathrm{path}}(\ell-1) \ge 8k$, then*

$$\lambda_{\min}(L(G_{m_{\mathrm{path}},\ell})_{-T}) \le b_\ell + \frac{24k}{m_{\mathrm{path}}(\ell-1)}. \tag{19}$$

3) *The constants satisfy $0 < b_\ell < a_\ell \le 1$, $b_\ell/a_\ell > c_*$ for every finite ℓ, and*

$$\lim_{\ell\to\infty} \frac{b_\ell}{a_\ell} = c_* := \left(\frac{2\arccos(1/3)}{\pi}\right)^2. \tag{20}$$

Proof. Put $h=\ell-1$. The quadratic form

$$q(y)=\sum_{j=1}^{h-1}(y_j-y_{j+1})^2+y_h^2$$

has smallest eigenvalue a_ℓ. Indeed its matrix is the path tridiagonal with first diagonal entry 1, all other diagonal entries 2, and off-diagonals -1 (for $h=1$ the matrix is $[1]$). Its eigenvectors have entries $\cos((j-\frac{1}{2})\vartheta)$, with $\vartheta=(2r-1)\pi/(2h+1)$, $r=1,\dots,h$; substitution gives the distinct eigenvalues $2-2\cos\vartheta$. Thus $q(y)\ge a_\ell\|y\|^2$.

When S is a cover, each component after grounding is either a star with $3m_{\text{path}}$ length-ℓ rays ending at ground, or an internal path with both endpoints grounded. For a star let x be its center value, $y^{(r)}$ the internal values of ray r, and write $Y=\sum_r\|y^{(r)}\|^2$. Its energy is

$$E=\sum_{r=1}^{3m_{\text{path}}}\left((x-y_1^{(r)})^2+q(y^{(r)})\right)\ge a_\ell Y.$$

Averaging $x^2\le 2(x-y_1^{(r)})^2+2(y_1^{(r)})^2$ over the rays yields $x^2\le 2(E+Y)/(3m_{\text{path}})$. Hence

$$x^2+Y\le\left(\frac{1+2/(3m_{\text{path}})}{a_\ell}+\frac{2}{3m_{\text{path}}}\right)E.$$

The Rayleigh principle gives (18). A path grounded at both ends has energy $q(y)+y_1^2\ge a_\ell\|y\|^2$, so the same lower bound holds for the whole matrix.

For the NO bound, choose an original uncovered edge uv. Give u,v value 1 and every other original vertex value 0. On every length-ℓ path with endpoint values $x,y\in\{0,1\}$, give its internal vertex j the value

$$z_j=\frac{x\sin((\ell-j)\theta_\ell)+y\sin(j\theta_\ell)}{\sin(\ell\theta_\ell)}. \tag{21}$$

These values satisfy $2z_j-z_{j-1}-z_{j+1}=b_\ell z_j$ at every internal vertex. At each of u,v, there is one original path type with endpoint values $(1,1)$ and two types with endpoint values $(1,0)$. The total outward flux, divided by m_{path}, is

$$3-\frac{3\sin((\ell-1)\theta_\ell)+\sin\theta_\ell}{\sin(\ell\theta_\ell)}=0,$$

by (17) and the sine subtraction formula. All other original vertices have value zero. If D is the sum of squared internal values across one copy of each original path type, then the full trial vector has squared norm $MD+2$ and energy exactly $b_\ell MD$.

All trial values lie in $[0,2]$. To see this, note that $\ell\theta_\ell/2<\ell\pi/(4\ell-2)\le\pi/3$ and $\ell\theta_\ell<\pi$. The $(1,1)$ path values equal

$$\frac{\cos((j-\ell/2)\theta_\ell)}{\cos(\ell\theta_\ell/2)}\in[1,2],$$

and the nonnegative $(1,0)$ and $(0,1)$ values are at most their sum. In particular the uv path alone gives $D\ge h$.

Now set the value of every internal vertex in T to zero. The original values already vanish on $T\cap V(H)$. Each altered internal vertex has degree 2, so at most $2k$ edges change energy. A changed edge has new energy at most 4, and the squared norm falls by at most $4k$. The modified vector is nonzero because $m_{\text{path}}h\ge 8k$ and it is an admissible Dirichlet test vector. Its Rayleigh quotient is at most

$$\frac{b_\ell MD+8k}{MD+2-4k}\le b_\ell+\frac{(8+4b_\ell)k}{MD-4k}\le b_\ell+\frac{24k}{Mh},$$

using $b_\ell<1$ and $MD\ge Mh\ge 8k$. This handles internal anchors without restricting the candidate set.

For completeness, the left side minus the right side of (17) equals 2 at zero and is negative at $\pi/(2\ell-1)$. Its derivative is negative in between, since $\sin((\ell-\frac{1}{2})\theta)\ge\sin(\theta/2)>0$ there and $3(\ell-\frac{1}{2})>1/2$. This proves existence, uniqueness, and $0<b_\ell<a_\ell\le 1$. Put $t_\ell=(\ell-\frac{1}{2})\theta_\ell\in(0,\pi/2)$. Equation (17) becomes $3\cos t_\ell=\cos(t_\ell/(2\ell-1))$. Its right side is strictly less than 1 and converges uniformly to 1, so $t_\ell>\arccos(1/3)$ and $t_\ell\to\arccos(1/3)$. Concavity of sine on $[0,\pi/2]$ gives

$$\frac{b_\ell}{a_\ell}=\left(\frac{\sin(t_\ell/(2\ell-1))}{\sin(\pi/(2(2\ell-1)))}\right)^2\ge\left(\frac{2t_\ell}{\pi}\right)^2>c_*.$$

Using $2-2\cos x=4\sin^2(x/2)$ and $\sin x/x\to 1$ proves (20). □

Theorem A10 (Amplified exact-budget spectral hardness)**.** *For every fixed $\gamma>c_*$, where c_* is defined in* (20)*, a polynomial-time γ-approximation for exact-budget grounded-$\lambda_{\min}$ maximization on connected simple unweighted graphs would imply* P = NP. *The reduction preserves the source budget k exactly and permits all graph vertices as candidates. In particular this excludes a same-budget spectral factor $1-1/e$. It also extends to a prescribed singleton seed and every fixed rational $\sigma\ge 0$ on connected positively weighted graphs. A randomized approximation with success probability at least $2/3$ would imply* NP = RP.

Proof. It suffices to consider $\gamma\le 1$. Choose a constant integer $\ell\ge 2$ with $b_\ell/a_\ell<\gamma$, possible by (20), and put $\Delta=\gamma a_\ell-b_\ell>0$. Fix an integer constant $C>\max\{8,32/\Delta\}$. Given a connected cubic vertex-cover instance (H,k), $1\le k<|V(H)|$, construct $G_{Ck,\ell}$ and keep budget k. It has $n+Ckm(\ell-1)$ vertices and $Ckm\ell$ edges, so its size is polynomial. Every vertex, including every path-interior vertex, is eligible for grounding. Define the constants

$$\alpha=\frac{a_\ell}{1+4/(3C)},\qquad\beta=b_\ell+\frac{24}{C(\ell-1)}.$$

Because $a_\ell\le 1$,

$$\gamma\alpha-\beta\ge\Delta-\frac{4\gamma a_\ell/3+24/(\ell-1)}{C}>\Delta-\frac{32}{C}>0.$$

If H has a cover of size at most k, enlarge it to size k and ground those original vertices. Equation (18) gives value at least α. If an arbitrary returned size-k set T has original part $T\cap V(H)$ that is not a cover, (19) gives value at most β. Thus a γ-approximation on a YES instance must return an original part that is a cover, since its value is at least $\gamma\alpha>\beta$. Check that original part combinatorially. On a NO instance no original part of size at most k can pass. This decides cubic vertex cover and proves the claimed hardness. The constants depend only

on the fixed target factor; strict inequalities allow fixed rational upper/lower enclosures when choosing them, independently of the input instance.

The special factor $1-1/e$ has an exact rational certificate. For $\ell = 4$, the ND path matrix and the sine recurrence give a_4 as the smallest positive zero of F, and $3F(b_4) = 1$, where $F(x) = 1-6x+5x^2-x^3$ is strictly decreasing on $[0,1/5]$. Direct rational evaluations give $F(99/500) > 0$, $F(1/5) < 0$, and $3F(309/2500) < 1$. Consequently

$$\frac{b_4}{a_4} < \frac{309/2500}{99/500} = \frac{103}{165} < \frac{5}{8} < 1 - \frac{1}{e}.$$

The last inequality follows from $e > 1+1+1/2+1/6+1/24 = 65/24 > 8/3$.

For a fixed seed and shift, multiply every path-edge weight by a constant positive integer W, add a new mandatory seed z, and join z to every other vertex by a unit-weight edge. After grounding z, any additional set T has matrix $WL(G_{Ck,\ell})_{-T} + (1+\sigma)I$. Choose W so that

$$W(\gamma\alpha - \beta) > (1-\gamma)(1+\sigma).$$

The YES lower bound $W\alpha + 1 + \sigma$ and the NO upper bound $W\beta + 1 + \sigma$ are then separated by factor γ. The graph and its rational encoding remain polynomial, and the budget still counts exactly k additions to the mandatory seed. The same combinatorial output check proves the reduction. For randomized algorithms this check has no false positives and succeeds with probability at least $2/3$ on YES instances, giving the NP = RP consequence: an NP-complete problem is decided with one-sided randomized error. □

APPENDIX C
MARGINAL AND HYBRID-STATE IDENTITIES

For the scalar identities, $A \succ 0$ is the current active matrix, $b = e_v$, $A_u = A + ubb^\top$, and $R_t = (A+tI)^{-1}$. Write $r_j(t) = b^\top R_t^j b$. The comparison matrix is $B = A + \beta bb^\top$ for a finite penalty and $B = A[-v,-v]$ for hard grounding. Thus $\Delta_f = \operatorname{tr} f(A) - \operatorname{tr} f(B)$.

Lemma A11 (Scalar resolvent marginal)**.** *For $t \geq 0$, $r_2(t) = -r_1'(t)$ and*

$$q_1(t) := \operatorname{tr}(A+tI)^{-1} - \operatorname{tr}(B+tI)^{-1} = \begin{cases} \dfrac{\beta r_2(t)}{1+\beta r_1(t)}, & \beta < \infty, \\ \dfrac{r_2(t)}{r_1(t)}, & \beta = \infty. \end{cases} \tag{22}$$

Consequently, for $0 < s < 1$,

$$\Delta_s = \frac{\sin(\pi s)}{\pi} \int_0^\infty t^{-s} q_1(t)\,dt, \qquad \Delta_1 = q_1(0). \tag{23}$$

Proof. Differentiating $(A+tI)R_t = I$ gives $R_t' = -R_t^2$, hence $r_1' = -r_2$. For the finite update, direct multiplication verifies the Sherman–Morrison identity

$$(A + tI + \beta bb^\top)^{-1} = R_t - \frac{\beta R_t bb^\top R_t}{1+\beta b^\top R_t b}.$$

Taking traces gives the first formula. For hard grounding reorder the candidate first and write R_t in blocks on $\{v\}, U$. The block inverse identity gives

$$(C+tI)^{-1} = (R_t)_{UU} - \frac{(R_t)_{Uv}(R_t)_{vU}}{(R_t)_{vv}}.$$

Its trace difference from R_t is $r_1 + \|(R_t)_{Uv}\|^2/r_1 = r_2/r_1$; the formula also holds when U is empty. The scalar identity

$$x^{-s} = \frac{\sin(\pi s)}{\pi} \int_0^\infty \frac{t^{-s}}{x+t}\,dt$$

follows from $t = xu$ and the beta integral $\int_0^\infty u^{-s}/(1+u)\,du = \Gamma(s)\Gamma(1-s) = \pi/\sin(\pi s)$. Diagonalization gives the corresponding trace identity. The integrals for each trace are finite, so subtracting them and using (22) proves (23). At $s = 1$ use the trace difference at $t = 0$ directly. □

Proposition A12 (Exact two-moment updates)**.** *At one shift let $R = (A+tI)^{-1}$, $x = Re_v$, $y = R^2 e_v$, and $s_x = r_{2,v} = x^\top x$. Put $c = (\beta^{-1} + r_{1,v})^{-1}$, using $\beta^{-1} = 0$ for hard grounding. Every remaining candidate u obeys*

$$r_{1,u}^+ = r_{1,u} - cx_u^2, \tag{24}$$

$$r_{2,u}^+ = r_{2,u} - 2cx_u y_u + c^2 s_x x_u^2. \tag{25}$$

Once x, y are available, updating all candidates at q shifts costs $O(nq)$ arithmetic and $O(nq)$ cache storage. Selected-column generation requires two solves per shift, or one retained Krylov basis from e_v with both small functions $(T+tI)^{-1}e_1, (T+tI)^{-2}e_1$.

Proof. Sherman–Morrison gives $R^+ = R - cxx^\top$ for soft grounding. For hard grounding the same matrix has zero row and column v; its complementary block is $(A[-v,-v]+tI)^{-1}$ by the block inverse identity in Lemma A11. Thus diagonal entries, and squared-column norms at remaining coordinates, can be computed before or after deleting the zero row and column. Squaring gives

$$(R - cxx^\top)^2 = R^2 - c(yx^\top + xy^\top) + c^2(x^\top x)xx^\top.$$

Taking diagonals proves both updates. Two shifted solves produce x and y; a retained Lanczos basis gives their projected versions from the two indicated small vectors, exactly at termination. Reconstructing both vectors at all shifts costs $O(nqh)$ after an h-step basis is stored, in addition to the large matrix–vector products. Only the cache update itself costs $O(nq)$. □

For the following all-order formula, R is the current zero-padded hard-grounded inverse, $x = Re_v$, and $c = 1/x_v$. The notation S includes the fixed seed, and $\bar{R}_v = R - cxx^\top$.

Proposition A13 (All-order independent decrements)**.** *Let $R = X_S^{-1}$ be zero padded on already selected coordinates, let v be active, $x = Re_v$, $c = 1/R_{vv}$, and $\bar{R}_v = R - cxx^\top$. For $i \geq 0$ define $a_i = R^i x$, $s_i = x^\top R^i x$, and, for each active u,*

$$\gamma_0(u) = a_0(u), \qquad \gamma_t(u) = a_t(u) - c\sum_{i=0}^{t-1} s_i \gamma_{t-1-i}(u).$$

Then $\gamma_t(u) = e_u^\top \bar{R}_v^t x \geq 0$ *and the moment decrement is*

$$D_j(u) = r_j(u,S) - r_j(u,S\cup\{v\}) = c\sum_{i=0}^{j-1} a_i(u)\gamma_{j-1-i}(u) \geq 0.$$

These decrements depend on selected columns, not on the cached candidate moments. After a last refresh at time b,

$$|\widehat{r}_j(u,S_t) - r_j(u,S_t)| \leq E_{j,u}^{(b)} + \sum_{i=b}^{t-1} \varepsilon_{j,u}^{(i)}, \qquad |\widehat{D}_j^{(i)} - D_j^{(i)}| \leq \varepsilon_{j,u}^{(i)}.$$

For exact columns all ε *are zero. In an implementation whose refresh flag changes only from false to true, whose refresh batch contains only false-flag candidates, and whose nonseed selections always have true flags, a run starting with all N candidate flags false has exactly* $N - n_{\mathrm{ref}}$ *never-refreshed candidates after* n_{ref} *total refreshes. All of them remain unselected.*

Proof. For small $|z_{\mathrm{ser}}|$, the rank-one inverse identity gives

$$(I - z_{\mathrm{ser}}\bar{R}_v)^{-1}x = \frac{(I - z_{\mathrm{ser}}R)^{-1}x}{1 + cz_{\mathrm{ser}}x^\top(I - z_{\mathrm{ser}}R)^{-1}x}.$$

Equating coefficients proves the recurrence for $\bar{R}_v^t x$. The telescoping identity

$$R^j - \bar{R}_v^j = \sum_{i=0}^{j-1} R^i(R - \bar{R}_v)\bar{R}_v^{j-1-i}$$

proves the displayed decrement. Grounded inverses and their zero-padded versions are entrywise nonnegative, so every displayed product is nonnegative. Subtracting the computed update from the true update and telescoping proves the error envelope; refreshing restarts it with that refresh's error budget. Finally each refresh changes one distinct false flag to true, so exactly n_{ref} flags have changed. Only true flags can be selected, proving the last assertion. □

We next give the maintained-solution sketch result with $A_0 = C^\top C$, $W = V \setminus S_0$, $N = |W|$, and R_T the zero-padded inverse.

Proposition A14 (Current-matrix sketch maintenance for hard CFCC)**.** *Let* $A_0 = X_{S_0} = C^\top C \succ 0$ *have dimension N. For* $T \subset W$*, write* R_T *for the zero-padded inverse of* $A_0[W\setminus T, W\setminus T]$*. Fix matrices* Π_C, Π_I *and maintain*

$$Y_1 = \Pi_C C R_T, \qquad Y_2 = \Pi_I R_T.$$

When grounding an active vertex v, one solve for $x = R_T e_v$ *updates both sketches exactly by*

$$Y_i' = Y_i - \frac{Y_i[:,v]x^\top}{x_v}, \qquad i = 1,2.$$

For every fixed T, if Π_C, Π_I *are independent Gaussian matrices with d rows and entry variance* $1/d$*, then* $\widehat{r}_1(u) = \|Y_1[:,u]\|^2$ *and* $\widehat{r}_2(u) = \|Y_2[:,u]\|^2$ *are unbiased estimates of the* current *moments* $r_1(u,T), r_2(u,T)$*.*

For $0 < \zeta < 1$*,* $0 < \delta < 1$*, and a maximum of* $k < N$ *new anchors, put* $\mathscr{N}_{N,k} = \sum_{t=0}^{k}\binom{N}{t}(N-t)$*. If*

$$d \geq 8\zeta^{-2}\log(4\mathscr{N}_{N,k}/\delta),$$

both estimates have simultaneous relative error at most ζ *at every active vertex of every such set, with probability at least* $1-\delta$*. This event covers adaptive choices made from the maintained sketches. Choosing the maximum estimated ratio* $\widehat{r}_2/\widehat{r}_1$*, or exactly rescoring a candidate pool containing that maximizer and choosing its best member, achieves a maximum-marginal fraction at least* $\alpha_{\mathrm{step}} = ((1-\zeta)/(1+\zeta))^2$*. For k selections its normalized gain is at least* $(1-e^{-\alpha_{\mathrm{step}}})\mathrm{OPT}_k$*. In particular* $\zeta \leq \varepsilon/4$ *gives the* $(1-e^{-1}-\varepsilon)$ *factor. The sufficient row count has order* $O(\zeta^{-2}((k+1)\log(N+1)+\log(1/\delta)))$*.*

Each update costs $O(Nd)$ *scalar work and uses* $O(Nd)$ *sketch storage. The selected-column solve can be reused from rescoring; no* $R_T^2 e_v$ *solve is needed for this sketch update.*

Moreover, for any row-vector error e supported on the active coordinates, its exact-column update $e' = e - e_v x^\top/x_v$ *obeys*

$$\|e'\|_{A_0}^2 = \|e\|_{A_0}^2 - e_v^2/x_v.$$

Thus accumulated initial sketch-solve errors are contractive in the fixed A_0 *energy norm under exact subsequent columns.*

Proof. Sherman–Morrison in its hard zero-padded form gives $R_{T+v} = R_T - xx^\top/x_v$. Multiplying on the left by $\Pi_C C$ or Π_I proves the update because $Y_i[:,v]$ is that left factor times x. On active coordinates $R_T A_0 R_T = R_T$, and all remaining rows and columns of both sides are zero. Consequently

$$\|CR_T e_u\|^2 = r_1(u,T), \qquad \|R_T e_u\|^2 = r_2(u,T).$$

For a fixed deterministic set and candidate, Gaussian norm concentration gives failure at most $2e^{-d\zeta^2/8}$ for each of the two vectors. There are $2\mathscr{N}_{N,k}$ vectors across all sets and candidates, so the stated row count gives the joint event. Since it holds simultaneously for this deterministic family, it also holds along every adaptive selection path. The bound $\mathscr{N}_{N,k} \leq N(k+1)N^k$ gives the stated asymptotic dimension.

On the event each ratio is between $(1-\zeta)/(1+\zeta)$ and $(1+\zeta)/(1-\zeta)$ times its true marginal. Comparing an estimated maximizer to a true maximizer therefore gives α_{step}. Exact rescoring a pool containing that candidate can only improve the true marginal at the same current set. The the integer analysis induction gives $1-e^{-\alpha_{\mathrm{step}}}$; since $1-\alpha_{\mathrm{step}} = 4\zeta/(1+\zeta)^2 \leq 4\zeta$, the final approximation follows. The two rank-one outer products and recomputation of column norms use $O(Nd)$ operations and storage, in addition to one selected-column solve.

For a supported row e, the active inverse equations imply $eA_0x = e_v$ and $x^\top A_0 x = x_v$. Expanding the squared energy norm gives the last identity. The new row has zero at v and at every previous anchor, so the contraction can be iterated. □

a) Initial solve errors in the maintained sketches.: For a matrix error $\mathscr{E}$ with supported rows, define $\|\mathscr{E}\|_{F,A_0}^2 = \mathrm{tr}(\mathscr{E}A_0\mathscr{E}^\top)$. Cauchy–Schwarz with the representer $R_T e_u$ gives $\|\mathscr{E}[:,u]\| \leq \sqrt{r_1(u,T)}\|\mathscr{E}\|_{F,A_0}$. Let $B \geq \lambda_{\max}(A_0)$ and let e_i bound the initial energy errors for the two solution sketches. The contraction above keeps those bounds valid under exact-

column updates. On the simultaneous Gaussian event, the relative squared-norm errors are at most

$$\zeta + 2\sqrt{1+\zeta}\, e_1 + e_1^2, \qquad \zeta + 2\sqrt{1+\zeta}\sqrt{B}\, e_2 + B e_2^2,$$

respectively. To obtain the second bound, use the spectral inequality $r_1 \le B r_2$. Both follow by expanding the perturbed norm square and bounding the cross term.

Appendix D
Adaptive Concentration and Capped Greedy

The first statement covers $J = p+1$ moments, and specializes to the two CFCC solution sketches for $p = 1$. The second gives the stopping-rule analysis for both general integer scores and the sharper CFCC ratio.

Theorem A15 (Uniform adaptive moment accuracy)**.** *Let all entries of Π_C, Π_I be independent centered Gaussians of variance $1/d$, or independent signs $\pm 1/\sqrt{d}$. For $J = p+1$, $0 < \zeta, \delta < 1$, put*

$$\mathscr{N}_{N,k} = \sum_{t=0}^{k} \binom{N}{t} (N-t), \qquad d_{\text{reuse}} = \left\lceil 8\zeta^{-2} \log \frac{2J\mathscr{N}_{N,k}}{\delta} \right\rceil.$$

With $d \ge d_{\text{reuse}}$ and exact sketch maintenance, all J moment estimates have relative error at most ζ at every active vertex of every $|T| \le k$, with probability at least $1-\delta$. The same event covers every adaptive selection rule, refresh pattern, and stopping decision. With $\zeta \le \eta/(16p)$, the computed integer marginals have relative error at most η by Theorem A20.

For comparison, new independent sketches in each of k rounds need only

$$d_{\text{fresh}} = \left\lceil 8\zeta^{-2} \log \frac{2JNk}{\delta} \right\rceil$$

for the same ideal-solve moment event along the chosen path. In particular, the sufficient dimensions have orders

$$d_{\text{reuse}} = O\big(p^2\eta^{-2}\left[(k+1)\log(N+1) + \log(p/\delta)\right]\big),$$
$$d_{\text{fresh}} = O\big(p^2\eta^{-2}\log(Npk/\delta)\big).$$

Proof. For a deterministic nonzero vector w, let Z be one unscaled sketch row dotted with $w/\|w\|$. In either distribution, $\mathbb{E}Z^2 = 1$, $\mathbb{E}Z^4 \le 3$, and $\mathbb{E}e^{tZ} \le e^{t^2/2}$. For signs the last inequality follows from $\cosh x \le e^{x^2/2}$ and independence. Integrating against an auxiliary standard Gaussian yields $\mathbb{E}e^{tZ^2} \le (1-2t)^{-1/2}$ for $0 < t < 1/2$. Chernoff's bound with $t = \zeta/[2(1+\zeta)]$ bounds the upper tail of the average of d independent Z^2 by

$$\exp\{-d[\zeta - \log(1+\zeta)]/2\} \le e^{-d\zeta^2/8}.$$

Here $\zeta - \log(1+\zeta) \ge \zeta^2/4$ on $[0,1]$, as follows by differentiating the difference. For the lower tail, $e^{-y} \le 1 - y + y^2/2$ for $y \ge 0$ gives $\mathbb{E}e^{-tZ^2} \le \exp(-t + 3t^2/2)$. Taking $t = \zeta/3$ gives $e^{-d\zeta^2/6}$. The combined two-sided failure is at most $2e^{-d\zeta^2/8}$.

The targets $R_T^{\ell} e_u$ and $CR_T^{\ell} e_u$ form a *deterministic* family with $J\mathscr{N}_{N,k}$ members when only the first J moments are retained. They are nonzero at active vertices. The preceding tail bound and a union bound prove the claim. Their correlations, including shared rows across powers, do not affect this union bound. Once the event holds, adaptively choosing a member of this family cannot invalidate it. The higher-power identity computes precisely these fixed-randomness sketches. Unbiasedness for each deterministic T supplies the moment identity, while this simultaneous event supplies the adaptive guarantee.

With fresh randomness, condition on the complete history at each round; its current matrix is then fixed independently of the new rows. A union bound over at most NJ current moments and a conditional failure budget δ/k proves the fresh formula. Finally $\mathscr{N}_{N,k} \le N(k+1)N^k$ and $\zeta = \eta/(16p)$ prove the stated orders. □

Theorem A16 (Hybrid retention and capped refreshes)**.** *At each state, suppose every not-yet-refreshed marginal estimate satisfies $(1-\eta)\Delta_u \le \widehat{\Delta}_u \le (1+\eta)\Delta_u$, and refreshed candidates are maintained exactly by the incremental analysis. Use the estimate itself as the heap key for an unrefreshed candidate and the exact marginal for a refreshed one. Let $0 \le \theta < 1$ be an acceptance tolerance. Process keys in descending order, exactly rescoring batches of unrefreshed candidates and retaining the best exact value b. Stop either when $b \ge (1-\theta)q_{\text{heap}}$, where q_{heap} is the largest remaining key, or after a positive refresh cap has been reached, and return the best processed candidate. Every processed refreshed candidate is included in this comparison. If a budget stop occurs, at least the initial maximum-key candidate has been processed. Then the selected marginal is at least*

$$\alpha_{\text{step}} \max_u \Delta_u, \qquad \alpha_{\text{step}} = \min\left\{ \frac{1-\eta}{1+\eta}, (1-\theta)(1-\eta) \right\}.$$

The k-round normalized gain is at least $(1 - e^{-\alpha_{\text{step}}})\mathrm{OPT}_k$. In particular $\eta \le \varepsilon/4$ and $\theta \le \varepsilon/2$ give $(1 - 1/e - \varepsilon)\mathrm{OPT}_k$ on the uniform event of Theorem A15.

For $p = 1$, put $\chi = (1+\zeta)/(1-\zeta)$ on its moment event. The sharper factor is $\alpha_{\text{step}} = \min\{\chi^{-2}, (1-\theta)\chi^{-1}\}$; thus $\zeta \le \varepsilon/8$ and $\theta \le \varepsilon/2$ suffice. Arbitrarily many candidates can be kept exact after their first refresh. Their total number of refreshes is $R \le N$.

Proof. Both types of key lie between $(1-\eta)\Delta_u$ and $(1+\eta)\Delta_u$. If u_* maximizes the true marginal and w has maximum initial key, then

$$\Delta_w \ge \frac{\text{key}(w)}{1+\eta} \ge \frac{1-\eta}{1+\eta}\Delta_{u_*}.$$

A budget stop has processed w and returns a candidate no worse than w. At an acceptance stop, every processed true value is at most b, and every unprocessed one is at most $q_{\text{heap}}/(1-\eta)$. Thus $b \ge (1-\theta)(1-\eta)\Delta_{u_*}$. Taking the minimum covers both stopping rules, including an empty remaining heap. Positive-budget batching may overshoot its nominal size without changing this argument.

By Theorem A2 the gain is normalized, monotone, and submodular relative to S_0. At each round its remaining gap to the best k-element extension is at most k times the largest current marginal. The gap therefore contracts by at least $1 - \alpha_{\text{step}}/k$ at

every step, proving $1-(1-\alpha_{\text{step}}/k)^k \geq 1-e^{-\alpha_{\text{step}}}$. Moreover $1-\alpha_{\text{step}} \leq \max\{2\eta, \eta+\theta\} \leq \varepsilon$, and $e^{-\alpha_{\text{step}}} - e^{-1} \leq 1-\alpha_{\text{step}}$ proves the claimed factor. For $p=1$ each ratio key is between Δ_u/χ and $\chi\Delta_u$. Repeat the same proof with these endpoints. Here $1-\chi^{-2} \leq 4\zeta$ and $1-(1-\theta)\chi^{-1} \leq \theta+2\zeta$. Exact retention follows from the independent-column decrement identity in the incremental analysis; each false-to-true refresh flag changes at most once. Replacing any sketched value by its exact value preserves all the preceding key inequalities, even for an adaptively chosen subset of candidates. □

APPENDIX E
NUMERICAL ERROR BUDGETS

In the following proposition the spectral bounds a, B refer to A_0, and the Frobenius energy norm is $\|\mathscr{E}\|_{F,A_0}^2 = \operatorname{tr}(\mathscr{E}A_0\mathscr{E}^\top)$. Zero padding is enforced on all grounded coordinates. The final paragraph applies to retained scalar entries and uses the independent decrements from Proposition A13.

Proposition A17 (Residual budgets for maintained sketches)**.** *Let $a \leq \lambda_{\min}(A_0)$ and $B \geq \lambda_{\max}(A_0)$. At a $p=1$ update, let $\widetilde{x}$ be supported on the current active set U, with $\widetilde{x}_v > 0$ and $\|A_0[U,U]\widetilde{x}_U - e_v\|_2 \leq \rho_{\text{res}}$. For either sketch, perform*

$$\widetilde{Y}' = \widetilde{Y} - \widetilde{Y}[:,v]\widetilde{x}^\top/\widetilde{x}_v + D,$$

where D has zero columns on all new and old anchors and $\|D\|_{F,A_0} \leq \omega$. If $e \geq \|\widetilde{Y}-Y\|_{F,A_0}$ before the update, then

$$e' = e + \frac{\|\widetilde{Y}[:,v]\|_2\rho_{\text{res}}}{\sqrt{a}\widetilde{x}_v} + \omega \tag{26}$$

is a valid bound afterwards. This recurrence is deterministic along adaptive selections. At every state the two moment errors are bounded by

$$\zeta + 2\sqrt{1+\zeta}\, e_1 + e_1^2, \qquad \zeta + 2\sqrt{1+\zeta}\sqrt{B}\, e_2 + Be_2^2$$

on the ideal-sketch event, where e_1, e_2 are their current accumulated energy bounds.

For higher powers, fix a left factor $\mathscr{F}$. A sufficient alternative certificate uses active-coordinate residuals $\rho_\ell = \|\widetilde{\mathscr{Y}_\ell}A_U - \widetilde{\mathscr{Y}_{\ell-1}}\|_F$, with $\widetilde{\mathscr{Y}_0} = \mathscr{F}[:,U]$ exact. Recursively let $E_0 = 0$ and $E_\ell = (E_{\ell-1}+\rho_\ell)/a$. Then $\|\widetilde{\mathscr{Y}_\ell} - \mathscr{F}[:,U]A_U^{-\ell}\|_F \leq E_\ell$. For even moments take $e = B^\ell E_\ell$, and for odd moments take $e = B^{\ell-1/2}E_\ell$. Their relative squared-norm error is at most $\zeta + 2\sqrt{1+\zeta}\, e + e^2$.

Proof. Let $x = R_T e_v$, $h = \widetilde{x} - x$, and $h_v = \widetilde{x}_v - x_v$. Active inverse equations give $\|h\|_{A_0} \leq \rho_{\text{res}}/\sqrt{a}$, $\langle h, x\rangle_{A_0} = h_v$, and $\|x\|_{A_0}^2 = x_v$. Therefore

$$\left\| \frac{\widetilde{x}}{\widetilde{x}_v} - \frac{x}{x_v} \right\|_{A_0}^2 = \frac{\|h\|_{A_0}^2 - h_v^2/x_v}{\widetilde{x}_v^2} \leq \frac{\rho_{\text{res}}^2}{a\widetilde{x}_v^2}.$$

Write $P = I - e_v x^\top/x_v$ and $E = \widetilde{Y} - Y$. Subtracting the exact update yields

$$\widetilde{Y}' - Y' = EP + \widetilde{Y}[:,v]\,(x/x_v - \widetilde{x}/\widetilde{x}_v)^\top + D.$$

Proposition A14 contracts EP in energy. The energy norm of an outer product is the product of the Euclidean norm of its left vector and the energy norm of its right vector. The triangle inequality proves (26). All new errors remain supported on unselected coordinates, so the argument iterates. Applying the coordinate-representer bounds after Proposition A14 proves the moment bounds.

For higher powers, subtract the exact chain equation at each level and multiply by A_U^{-1}, whose norm is at most $1/a$ by principal-submatrix interlacing. This proves the E_ℓ recurrence. The true target squared norms are at least $B^{-2\ell}$ and $B^{-(2\ell-1)}$, respectively. On the ideal JL event, their sketched norm is at most $\sqrt{1+\zeta}$ times the true target norm. Expanding a perturbed squared norm and bounding its cross term proves the final error expression. □

a) Instantiating numerical budgets.: For example, a $p=1$ target moment tolerance $\bar{\zeta} < 1$ is met by an ideal-sketch tolerance $\zeta = \bar{\zeta}/2$ and $e_1, \sqrt{B}e_2 \leq \bar{\zeta}/8$; substitution in the preceding bound is at most $\bar{\zeta}$. Initial solve errors, subsequent column residuals, and rounding terms share these budgets. The same substitution applies to the higher-power normalized errors e. Refreshed scalar entries use the the incremental analysis decrement intervals and refresh-error bounds; if their resulting relative marginal error is at most ν, replacing exact values throughout the hybrid proof loses at most the additional factor $(1-\nu)/(1+\nu)$, with key tolerance enlarged to $\max\{\eta, \nu\}$. This follows by comparing the largest computed processed value with the largest true processed value before applying the two stopping arguments. Residual evaluation for every higher-power level costs $O(dL(m+N))$ per checked state and is charged separately when used. These statements specify numerical conditions, in addition to the sketch event, for an inexact implementation of the preceding algorithms.

b) Scalar decrement errors: Let a selected-column chain have true-residual bounds ρ_i and spectral lower bound $a_* > 0$. With $a_{-1} = e_v$, its Euclidean errors satisfy $E_{-1} = 0$, $E_i \leq (E_{i-1} + \rho_i)/a_*$. Use component intervals $\widehat{a}_i(u) \pm E_i$, and intervals for s_i centered at $\widehat{a}_0^\top \widehat{a}_i$ with radius $\|\widehat{a}_0\|E_i + \|\widehat{a}_i\|E_0 + E_0E_i$. The reciprocal of a positive interval for $a_0(v)$ encloses c. Propagating these intervals through the finite γ_t, D_j recurrences encloses every true decrement by induction. Their distance from the computed decrement gives the ε above. This construction has the same $O(NJ^2)$ scalar order as the update. The residual inequality follows by applying R to each chain equation; the inner-product radius follows by expansion and Cauchy–Schwarz.

APPENDIX F
SEED-PRECONDITIONER REUSE

Write $W = V \setminus S_0$, $N = |W|$, $A_0 = X_{S_0} \succ 0$, $T = S \setminus S_0$, $t = |T|$, and $U = W \setminus T$. All inverses in this subsection act on their active coordinates. An inverse preconditioner $B_0 \succ 0$ for A_0 is applied to a vector on U by zero extension to W, application of B_0, and restriction back to U. Its active matrix is $B_U = B_0[U,U]$.

Theorem A18 (Masked seed preconditioning)**.** *Suppose $aA_0^{-1} \preceq B_0 \preceq bA_0^{-1}$ for $0 < a \leq b$. Let $A = A_0[U,U]$ and $D_U = |U| > 0$. Every eigenvalue of $P_U = A^{1/2}B_UA^{1/2}$ is at*

least a, and at most t of its eigenvalues exceed b. Let $n_{\mathrm{out}} \le \min\{t, D_U\}$ count those exceeding b, and set $\chi_c = b/a$. In exact arithmetic PCG obeys, for $h \ge n_{\mathrm{out}}$ and $\chi_c > 1$,

$$\frac{\|e_h\|_A}{\|e_0\|_A} \le 2\left(\frac{\sqrt{\chi_c}-1}{\sqrt{\chi_c}+1}\right)^{h-n_{\mathrm{out}}}.$$

It therefore reaches relative energy error ξ in at most

$$\min\left\{D_U,\ t + \left\lceil \frac{\log(2/\xi)}{\log((\sqrt{\chi_c}+1)/(\sqrt{\chi_c}-1))} \right\rceil\right\}$$

iterations. For $a = b$, termination occurs within $\min\{D_U, t+1\}$ iterations. These bounds hold simultaneously for every T on the single seed spectral-approximation event.

The identity-padded matrix $\overline{A} = A \oplus I_T$ differs from A_0 by rank at most $2t$.

Proof. Order U before T and write $A_0 = \left(\begin{smallmatrix} A & C \\ C^\top & D \end{smallmatrix}\right)$. The block inverse formula gives

$$(A_0^{-1})[U,U] = A^{-1} + A^{-1}C(D - C^\top A^{-1} C)^{-1} C^\top A^{-1}.$$

The Schur complement is positive definite. Congruence by $A^{1/2}$ thus yields $\mathscr{K} = I + ZZ^\top$ with $\operatorname{rank} Z \le t$. Compression preserves the assumed Loewner inequalities, so $a\mathscr{K} \preceq P_U \preceq b\mathscr{K}$. In particular $P_U \succeq aI$. On the subspace $\ker Z^\top$, of codimension at most t, the Rayleigh quotient of P_U is at most b. The min–max principle proves the outlier count. The preconditioned system $B_U^{1/2} A B_U^{1/2}$ has the same spectrum as P_U.

For completeness, CG minimizes the energy norm of the error over its affine Krylov space: its conjugate search directions are pairwise orthogonal in the system inner product and exact line minimization makes the final error orthogonal to their span. Apply this property to $P_{\mathrm{pcg}} = B_U^{1/2} A B_U^{1/2}$ and the transformed error $\widetilde{e} = B_U^{-1/2} e$, for which $\|\widetilde{e}\|_{P_{\mathrm{pcg}}} = \|e\|_A$. The Krylov error candidates are $P(P_{\mathrm{pcg}})\widetilde{e}_0$ for polynomials of degree at most h with $P(0) = 1$. Diagonalizing P_{pcg} bounds their P_{pcg}-norm by $\max_{\lambda \in \mathrm{spec}(P_{\mathrm{pcg}})} |P(\lambda)| \|e_0\|_A$. Since P_{pcg} and P_U have the same spectrum, this is the required polynomial bound in terms of P_U.

Let $\mu_1, \dots, \mu_{n_{\mathrm{out}}} > b$ be the outliers, with multiplicity. Multiply the degree-$(h - n_{\mathrm{out}})$ Chebyshev residual polynomial on $[a,b]$, normalized to equal one at zero, by $\prod_{i=1}^{n_{\mathrm{out}}} (1 - z/\mu_i)$. The product vanishes at every outlier and has absolute value at most one on $[a,b]$. The Chebyshev factor is bounded by $2((\sqrt{\chi_c}-1)/(\sqrt{\chi_c}+1))^{h-n_{\mathrm{out}}}$ there, proving the result. For $a = b$ the polynomial $(1 - z/a)\prod_i (1 - z/\mu_i)$ annihilates the spectrum. The usual degree-D_U annihilating polynomial gives the dimension cap. Finally

$$\overline{A} - A_0 = \begin{pmatrix} 0 & -C \\ -C^\top & I - D \end{pmatrix} = \begin{pmatrix} 0 & -C \\ 0 & I - D \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ -C^\top & 0 \end{pmatrix},$$

a sum of two matrices of rank at most t. The stronger one-sided outlier conclusion follows from the block inverse calculation, rather than from this rank count alone. Every argument is deterministic for every principal subset once B_0 satisfies its initial spectral inequalities. □

a) Approximate Cholesky instantiation.: Contract the seed to a vertex g, and represent any diagonal killing by edges to g. Let $\mathscr{L}$ be the resulting connected Laplacian, whose principal matrix off g is A_0. The SparseCholesky theorem of Kyng–Sachdeva [3], Theorem 3.1, supplies $\widetilde{\mathscr{L}}$ with $(1-\varepsilon_c)\mathscr{L} \preceq \widetilde{\mathscr{L}} \preceq (1+\varepsilon_c)\mathscr{L}$ except with probability δ_c. Its expected construction work and factor storage are $O(\varepsilon_c^{-2}(m_E + N)\log^2((N+1)/\delta_c)\log(N+1))$; Appendix B gives the high-probability work variant with the stated sampling rule. The sparse inverse application has work $O(F+N)$, where F is the stored factor size.

Define Q to have rows $e_u^\top - e_g^\top$, $u \in W$. Then $B_0 = Q\widetilde{\mathscr{L}}^{+} Q^\top$ equals the inverse of $\widetilde{\mathscr{L}}[W,W]$: solving with balanced source $(r, -\mathbf{1}^\top r)$ and subtracting the ground potential solves the grounded system. Restriction of the Laplacian spectral inequalities and inversion give $a = (1+\varepsilon_c)^{-1}$, $b = (1-\varepsilon_c)^{-1}$, hence $\chi_c = (1+\varepsilon_c)/(1-\varepsilon_c)$. At $\varepsilon_c = 1/2$, $\chi_c = 3$ and the iteration bound is $t + O(\log(1/\xi))$. This instantiation uses the imported factorization theorem; Theorem A18 proves its simultaneous reuse under grounding.

Appendix G
Integer Marginals and Higher-Power Sketches

These results prove Proposition 5 and develop the higher-power maintained-sketch extension. In the first two statements, $A \succ 0$, $b = e_v$, $r_j = b^\top (A + tI)^{-j} b$, and $\beta^{-1} = 0$ denotes the hard case. The logarithmic recurrence is given explicitly in Proposition A29 below.

Proposition A19 (Positive integer marginal representation)**.** *At any nonnegative shift let $R = (A + tI)^{-1}$, $x = Rb$, $c = (\beta^{-1} + r_1)^{-1}$, using $\beta^{-1} = 0$ for hard grounding, and $\bar{R}_v = R - cxx^\top \succeq 0$. Define*

$$a_j = c r_{j+1}, \qquad d_j = c x^\top \bar{R}_v^{j-1} x \quad (j \ge 1),$$
$$H_0 = 1, \qquad H_j = \sum_{i=1}^{j} d_i H_{j-i}.$$

The integer marginal admits the noncancelling formula

$$q_p(t) = \sum_{j=1}^{p} j d_j H_{p-j}, \qquad a_p \le q_p(t) \le p a_p. \tag{27}$$

All terms are nonnegative. In particular, with ν the probability measure defined below,

$$q_2 = 2a_2 - a_1^2 = a_2 + \mathrm{Var}_\nu(Z), \qquad a_1^2 \le a_2. \tag{28}$$

Proof. For soft grounding $\bar{R}_v = (A + tI + \beta bb^\top)^{-1}$. For hard grounding it is the zero-padded inverse of the deleted principal matrix. These facts follow from the principal-inverse identities, so $q_p = \operatorname{tr} R^p - \operatorname{tr} \bar{R}_v^p$ and $\bar{R}_v \succeq 0$ in both cases. Write $D(z_{\mathrm{ser}}) = \sum_{j \ge 1} d_j z_{\mathrm{ser}}^j = c z_{\mathrm{ser}} x^\top (I - z_{\mathrm{ser}} \bar{R}_v)^{-1} x$. For sufficiently small $|z_{\mathrm{ser}}|$, the determinant lemma gives

$$\frac{\det(I - z_{\mathrm{ser}} \bar{R}_v)}{\det(I - z_{\mathrm{ser}} R)} = \frac{1}{1 - D(z_{\mathrm{ser}})},$$
$$\sum_{p \ge 1} \frac{q_p}{p} z_{\mathrm{ser}}^p = -\log(1 - D(z_{\mathrm{ser}})).$$

Every d_j is nonnegative by positive semidefiniteness of $\bar{R}_v$. The reciprocal power series has coefficients H_j satisfying the stated positive recurrence. Differentiating the logarithmic series proves (27)'s equality.

Alternatively expand words in $R = \bar{R}_v + cxx^\top$ inside $\operatorname{tr} R^p$. Marking one of the ℓ occurrences of $cxx^\top$ and cyclically rotating the trace gives

$$q_p = \sum_{\ell=1}^{p} \frac{p}{\ell} \sum_{\substack{j_1+\cdots+j_\ell=p \\ j_i\geq 1}} d_{j_1}\cdots d_{j_\ell}.$$

This counting identity follows because there are p possible positions for a marked occurrence and each word is counted ℓ times. The same expansion in $cx^\top R^{p-1}x = cr_{p+1} = a_p$ gives the identical sum without the factor p/ℓ. Every factor is nonnegative and $1 \leq p/\ell \leq p$, proving both magnitude bounds.

For the variance form diagonalize R, with eigenvalues $\lambda_i(R) > 0$ and spectral weights w_i of b. Put mass $cw_i\lambda_i(R)$ at $\lambda_i(R)$ and mass $1 - cr_1$ at zero. These masses sum to one; in the soft case the zero mass equals $\beta^{-1}/(\beta^{-1}+r_1)$, and it is zero in the hard case. The resulting distribution ν has $\mathbb{E}_\nu Z^j = a_j$. Thus $a_1^2 \leq a_2$, and the formula for q_2 from the scalar identities becomes (28). The subtracted term is at most half of $2a_2$, so second-order cancellation is uniformly bounded, independently of dimension and condition number. □

Theorem A20 (Polynomial-precision integer marginal estimation)**.** *Use the notation of Proposition A19, at any fixed shift. If $|\widehat{r}_j/r_j - 1| \leq \zeta < 1$ for $j = 1,\ldots,p+1$, the scalar logarithmic recurrence satisfies*

$$\left|\frac{\widehat{q}_p}{q_p} - 1\right| \leq \left(1 + \frac{4\zeta}{1-\zeta}\right)^p - 1. \tag{29}$$

For $0 < \eta \leq 1$, choosing $\zeta \leq \eta/(16p)$ makes this error at most η. The assertion holds for soft and hard grounding, independently of dimension and condition number.

Proof. Work with formal power series through degree p. Set

$$F(z_{\mathrm{ser}}) = 1 + \sum_{j\geq 1} a_j z_{\mathrm{ser}}^j, \qquad D(z_{\mathrm{ser}}) = \sum_{j\geq 1} d_j z_{\mathrm{ser}}^j.$$

The determinant identities in Proposition A19 give $F^{-1} = 1 - D$, with $a_j, d_j \geq 0$, and hence

$$a_j = d_j + \sum_{i=1}^{j-1} a_i d_{j-i}. \tag{30}$$

Also $q_p = p[z_{\mathrm{ser}}^p]\log F$, $q_p \geq a_p$, and Holder's inequality for the probability measure constructed there gives $a_{j_1}\cdots a_{j_\ell} \leq a_p$ whenever $j_1+\cdots+j_\ell = p$ and $j_i \geq 1$.

Put $\widehat{a}_j = \widehat{r}_{j+1}/(\beta^{-1}+\widehat{r}_1)$. Positivity of the exact penalty yields

$$\frac{1-\zeta}{1+\zeta} \leq \frac{\widehat{a}_j}{a_j} \leq \frac{1+\zeta}{1-\zeta}, \qquad |\widehat{a}_j - a_j| \leq \alpha a_j, \quad \alpha = \frac{2\zeta}{1-\zeta}.$$

For $E(z_{\mathrm{ser}}) = \sum_{j=1}^{p}(\widehat{a}_j - a_j)z_{\mathrm{ser}}^j$, define $H = E/F = E(1-D)$. For $1 \leq j \leq p$, identity (30) implies

$$|[z_{\mathrm{ser}}^j]H| \leq \alpha\left(a_j + \sum_{i=1}^{j-1} a_i d_{j-i}\right) = \alpha(2a_j - d_j) \leq 2\alpha a_j.$$

The formal identity $\log(F+E) - \log F = \log(1+H)$ then gives

$$\begin{aligned}|\widehat{q}_p - q_p| &\leq p\sum_{\ell=1}^{p}\frac{(2\alpha)^\ell}{\ell}\sum_{\substack{j_1+\cdots+j_\ell=p\\ j_i\geq 1}} a_{j_1}\cdots a_{j_\ell} \\ &\leq a_p\sum_{\ell=1}^{p}\binom{p}{\ell}(2\alpha)^\ell = a_p((1+2\alpha)^p - 1).\end{aligned}$$

Division by $q_p \geq a_p$ proves (29). All identities concern finitely many coefficients, so convergence of a series or realizability of the perturbed moments is unnecessary. Finally $\zeta \leq \eta/(16p)$ implies $4p\zeta/(1-\zeta) \leq 4\eta/15$. Thus the bound is at most $e^{4\eta/15} - 1 \leq \eta$: convexity bounds it by $\eta(e^{4/15}-1)$ on $[0,1]$, and $e^{4/15} \leq (1-4/15)^{-1} = 15/11 < 2$. □

Proposition A21 (Higher-power sketch maintenance)**.** *Fix a matrix $\mathscr{F}$ and an integer $\ell_{\max} \geq 1$, and store $\mathscr{Y}_\ell = \mathscr{F}R_T^\ell$ for $1 \leq \ell \leq \ell_{\max}$. Before grounding v, put $R = R_T$, $x = Re_v$, $c = 1/x_v$, $a_i = R^i x$, and $s_i = x^\top a_i$. Define*

$$b_0 = x, \qquad b_t = a_t - c\sum_{i=0}^{t-1} s_i b_{t-1-i} \quad (1 \leq t < \ell_{\max}).$$

All levels update simultaneously by

$$\mathscr{Y}_\ell' = \mathscr{Y}_\ell - c\sum_{i=0}^{\ell-1}\mathscr{Y}_{i+1}[:,v]b_{\ell-1-i}^\top. \tag{31}$$

Every vector on the right is its pre-update value. Apply this construction to $\mathscr{F} = \Pi_I$ and $\mathscr{F} = \Pi_C C$, with d rows each. The squared columns of $\Pi_I R_T^\ell$ and $\Pi_C C R_T^\ell$ estimate, respectively, $r_{2\ell}(u,T)$ and $r_{2\ell-1}(u,T)$. Thus moments through $p+1$ need $\ell_{\max} = \lceil (p+1)/2 \rceil$.

Initialization uses at most $2d\ell_{\max}$ right-hand-side solves. An update uses at most $\ell_{\max}$ selected-column solves, $O(Nd\ell_{\max}^2)$ arithmetic, and $O(Nd\ell_{\max})$ storage; existing selected columns can be reused. The formula specializes to the two rank-one updates of Proposition A14 when $p = 1$. For $p = 2$, it additionally gives

$$\mathscr{Y}_2' = \mathscr{Y}_2 - c\mathscr{Y}_1[:,v](Rx - c\|x\|^2 x)^\top - c\mathscr{Y}_2[:,v]x^\top.$$

Proof. Let $\bar{R}_v = R - cxx^\top = R_{T+v}$. The generating-function calculation in the incremental analysis gives $b_t = \bar{R}_v^t x$; alternatively it follows by comparing coefficients in

$$(I - z_{\mathrm{ser}}\bar{R}_v)^{-1}x = \frac{(I - z_{\mathrm{ser}}R)^{-1}x}{1 + cz_{\mathrm{ser}}x^\top(I - z_{\mathrm{ser}}R)^{-1}x}.$$

The noncommutative telescoping identity

$$R^\ell - \bar{R}_v^\ell = c\sum_{i=0}^{\ell-1} R^i x x^\top \bar{R}_v^{\ell-1-i}$$

follows by expanding the consecutive differences $R^{i+1}\bar{R}_v^{\ell-1-i} - R^i\bar{R}_v^{\ell-i}$ and summing. Multiplication by $\mathscr{F}$ proves (31), since

$\mathscr{F}R^i x = \mathscr{Y}_{i+1}[:,v]$ and $\bar{R}_v$ is symmetric. Snapshotting all columns $\mathscr{Y}_\ell[:,v]$ before any update gives an in-place implementation of this identity.

For every T, $R_T A_0 R_T = R_T$. Therefore

$$\|R_T^\ell e_u\|^2 = r_{2\ell}(u,T), \qquad \|CR_T^\ell e_u\|^2 = r_{2\ell-1}(u,T).$$

These identities prove the moment targets for isotropic sketches. The chain $a_0,\dots,a_{\ell_{\max}-1}$ uses $\ell_{\max}$ solves; its inner products and the b_t recurrence cost $O(N\ell_{\max}^2)$. There are $\sum_{\ell=1}^{\ell_{\max}} \ell = O(\ell_{\max}^2)$ outer products per sketch family, costing $O(Nd\ell_{\max}^2)$. The stored levels and selected chains take $O(Nd\ell_{\max})$ space. Repeatedly solving each of the two families of d starting rows gives initialization. For $\ell_{\max}=2$, $b_1 = Rx - c\|x\|^2 x$, giving the final formula. □

Appendix H
Spectral Translation and Finite Removal

All spectral comparisons here leave a nonempty grounded matrix. The common seed and budget fix the comparison family. For the bicriteria statement, the certified lazy rule is described in Appendix I; its step factor is $1-\theta$. The notation X_S denotes the active grounded matrix A_S^∞.

Theorem A22 (Uniform inverse-power spectral rate)**.** *Let $X \succ 0$ have dimension D and ordered eigenvalues $\lambda_1 \le \cdots \le \lambda_D$. For every $s > 0$,*

$$\lambda_1^{-1} \le (\operatorname{tr} X^{-s})^{1/s} \le D^{1/s}\lambda_1^{-1}. \tag{32}$$

The relative excess is at most $e^{(\log D)/s} - 1$; it is at most η whenever $s \ge \log D/\log(1+\eta)$. If λ_1 has multiplicity $r < D$ and $\rho_{\text{spec}} = \lambda_1/\lambda_{r+1} < 1$, then

$$0 \le \log\big(\lambda_1(\operatorname{tr} X^{-s})^{1/s}\big) - \frac{\log r}{s} \le \frac{D-r}{rs}\rho_{\text{spec}}^s.$$

At a fixed cardinality and matrix dimension D, any set whose power-trace objective is within a multiplicative factor $\alpha_{\text{cost}} \ge 1$ of its minimum satisfies

$$\lambda_{\min}(X_{\text{out}}) \ge (\alpha_{\text{cost}} D)^{-1/s}\lambda_k^*, \tag{33}$$

where λ_k^* is the maximum smallest eigenvalue at that cardinality and the same seed. In particular an exact power-trace minimizer has factor $D^{-1/s}$.

Proof. Factor the smallest eigenvalue out of the trace:

$$(\operatorname{tr} X^{-s})^{1/s} = \lambda_1^{-1}\left(\sum_{i=1}^{D}(\lambda_1/\lambda_i)^s\right)^{1/s}.$$

The sum lies between 1 and D, giving (32) and its stated rate. With multiplicity r it lies between r and $r+(D-r)\rho_{\text{spec}}^s$. Taking logarithms and using $\log(1+x) \le x$ proves the refined inequality. If $r = D$, the sum equals D exactly.

Let O_λ maximize the smallest eigenvalue and O_s minimize the trace, at the same budget. Then

$$\begin{aligned}\lambda_{\min}(X_{\text{out}})^{-s} &\le \Phi_s(\text{out}) \le \alpha_{\text{cost}}\Phi_s(O_s)\\ &\le \alpha_{\text{cost}}\Phi_s(O_\lambda) \le \alpha_{\text{cost}} D(\lambda_k^*)^{-s}.\end{aligned}$$

Taking the inverse sth root proves (33). The scalar root is used to compare minimizers and spectral values; the greedy gain remains defined using $\Phi_s = \operatorname{tr} X^{-s}$. □

Theorem A23 (Baseline-aware translation of the gain guarantee)**.** *Fix $s > 0$, a seed, and k additional anchors. Put $\Phi_s^0 = \Phi_s(S_0)$, $\Psi_s^* = \min_{|U|=k}\Phi_s(S_0 \cup U)$, and $C_s = \Phi_s^0/\Psi_s^*$. Let D be the common final dimension and λ_k^* the optimal smallest eigenvalue. If a returned set satisfies a gain factor $\gamma \in (0,1]$, put $\rho_{\text{gain}} = 1-\gamma$. Then*

$$\begin{aligned}\frac{\lambda_{\min}(X_{\text{out}})}{\lambda_k^*} &\ge \big[\rho_{\text{gain}}\Phi_s^0(\lambda_k^*)^s + \gamma D\big]^{-1/s}\\ &\ge \big[D(\rho_{\text{gain}}C_s + \gamma)\big]^{-1/s}.\end{aligned} \tag{34}$$

Thus a contrast bound $C_s \le C$ at the chosen exponent and

$$s \ge \frac{\log(D(\rho_{\text{gain}}C + \gamma))}{\log(1+\eta)}$$

certify a factor $1/(1+\eta)$ on the smallest eigenvalue. Apply this bound on the success event of any of the stated gain guarantees, with its corresponding γ. Without a baseline-contrast condition, any fixed gain factor $\gamma < 1$ alone admits arbitrarily small spectral ratios, even on a connected weighted four-vertex star and at every integer exponent $s \ge 1$.

Proof. The gain guarantee is exactly the residual-objective inequality

$$\Phi_s(\text{out}) \le \rho_{\text{gain}}\Phi_s^0 + \gamma\Psi_s^*.$$

Since $\Psi_s^* \le D(\lambda_k^*)^{-s}$ and $\lambda_{\min}(X_{\text{out}})^{-s} \le \Phi_s(\text{out})$, the first spectral bound follows. Also $\Phi_s^0(\lambda_k^*)^s = C_s\Psi_s^*(\lambda_k^*)^s \le C_s D$, giving the second bound and the stated sufficient exponent. The bound holds on the corresponding gain event.

For the final assertion, take a star with center 0, leaves 1,2,3, and respective edge weights $b/2, b, 1$, where $0 < b < 1$. The fixed seed is $S_0 = \{0\}$, so the current matrix is $\operatorname{diag}(b/2, b, 1)$. Two additional anchors are allowed. The optimal choice $\{1,2\}$ leaves the eigenvalue 1; the choice $\{1,3\}$ leaves the eigenvalue b. Their gain ratio is exactly

$$\frac{(2/b)^s + 1}{(2/b)^s + b^{-s}} = \frac{2^s + b^s}{2^s + 1} \ge \frac{2}{3} > 1 - e^{-1} \qquad (s \ge 1).$$

Nevertheless the spectral ratio is b, which tends to zero. For a larger prescribed $\gamma < 1$, replace the weight $b/2$ by b/m_{amp} with integer $m_{\text{amp}} \ge \max\{2, \gamma/(1-\gamma)\}$. The gain ratio is then $(m_{\text{amp}}^s + b^s)/(m_{\text{amp}}^s + 1) \ge \gamma$ for every integer $s \ge 1$, and the spectral ratio is still b. This is a counterexample to an implication from the gain factor alone: it distinguishes that factor from the multiplicative trace condition in (33). For fixed $\rho_{\text{gain}} > 0$ and $\lambda_k^* > \lambda_{\min}(X_{S_0})$, the first bound in (34) tends to $\lambda_{\min}(X_{S_0})/\lambda_k^*$ as $s \to \infty$, by Theorem A22. Increasing the exponent makes the surrogate sharper while the baseline term remains explicit. □

Corollary A24 (A spectral bicriteria guarantee)**.** *Compare with the optimal set of k additional anchors, with remaining dimension D_k. Let the initial matrix have dimension D_0, let*

$a > 0$ lower-bound its smallest eigenvalue, and let B upper-bound λ_k^*. Run the same certified lazy rule with step factor $1-\theta$ for $\ell \geq k$ rounds. Whenever the budget is feasible and hard grounding leaves a vertex,

$$\Phi_s(S_\ell) \leq e^{-(1-\theta)\ell/k}\Phi_s^0 + \Psi_{s,k}^*. \tag{35}$$

For a target $\eta > 0$, choose an integer

$$\begin{aligned} s &\geq \left\lceil \frac{\log(2D_k)}{\log(1+\eta)} \right\rceil, \\ \ell &\geq \max\left\{k, \left\lceil \frac{k}{1-\theta}\left(\log\frac{D_0}{D_k} + s\log\frac{B}{a}\right)\right\rceil\right\}. \end{aligned} \tag{36}$$

Then, on the same probability-$1-\delta$ event, $\lambda_{\min}(X_{S_\ell}) \geq \lambda_k^*/(1+\eta)$.

Proof. The maximum-marginal argument in Theorem A16 may compare every round with a fixed size-k optimum, even after more than k rounds. Its residual recurrence is

$$\Phi_s(S_{j+1}) - \Psi_{s,k}^* \leq \left(1 - \frac{1-\theta}{k}\right)(\Phi_s(S_j) - \Psi_{s,k}^*).$$

If the parenthesized gap is negative, monotonicity also implies this inequality. Iteration and the exponential bound prove (35). Now $\Phi_s^0 \leq D_0 a^{-s}$ and $\Psi_{s,k}^* \leq D_k(\lambda_k^*)^{-s}$. The second choice in (36) bounds the first term in (35) by $D_k B^{-s} \leq D_k(\lambda_k^*)^{-s}$. Therefore

$$\lambda_{\min}(X_{S_\ell})^{-s} \leq \Phi_s(S_\ell) \leq 2D_k(\lambda_k^*)^{-s}.$$

The first choice of exponent proves the spectral factor. The permanent-key argument covers all ℓ rounds on its initial event. □

Proposition A25 (Finite-removal expansion and spectral weighting)**.** *Let X be a grounded Laplacian with a simple smallest eigenpair (λ_1, ϕ_1), $\|\phi_1\| = 1$, and orthonormal eigenbasis ϕ_i. For an active candidate v, let $R = X^{-1}$, $x = Re_v$, $c = 1/r_1$, $D = cxx^\top$, and $a_j = cr_{j+1}$. For $0 \leq \vartheta \leq 1$ put $Q_p(\vartheta) = \operatorname{tr} R^p - \operatorname{tr}(R - \vartheta D)^p$. Its endpoint $Q_p(1)$ is the hard-grounding marginal. For $p = 2,3,4$,*

$$\begin{aligned} Q_2(\vartheta) &= 2\vartheta a_2 - \vartheta^2 a_1^2, \\ Q_3(\vartheta) &= 3\vartheta a_3 - 3\vartheta^2 a_1 a_2 + \vartheta^3 a_1^3, \\ Q_4(\vartheta) &= 4\vartheta a_4 - \vartheta^2(4a_1a_3 + 2a_2^2) + 4\vartheta^3 a_1^2 a_2 - \vartheta^4 a_1^4. \end{aligned}$$

Their first-order coefficient satisfies

$$Q_p'(0) = \frac{p}{r_1(v)} \sum_i \lambda_i^{-(p+1)} \phi_i(v)^2.$$

For the weighted Zhou score $h_v = 2\phi_1(v)\sum_{w \neq v} w_{vw}\phi_1(w)$,

$$h_v = 2(X_{vv} - \lambda_1)\phi_1(v)^2.$$

Whenever $X_{vv} > \lambda_1$,

$$Q_p'(0) = \frac{p\lambda_1^{-(p+1)} h_v}{2r_1(v)(X_{vv} - \lambda_1)} + \frac{p}{r_1(v)} \sum_{i \geq 2} \lambda_i^{-(p+1)} \phi_i(v)^2.$$

The tail sum is at most $(1-\phi_1(v)^2)\lambda_2^{-(p+1)}$. For every integer $p \geq 2$,

$$0 \leq \vartheta Q_p'(0) - Q_p(\vartheta) \leq \frac{p(p-1)}{2}\vartheta^2 a_1^2 \|R\|^{p-2}.$$

For an infinitesimal soft penalty the coefficient is

$$p\sum_i \lambda_i^{-(p+1)} \phi_i(v)^2.$$

Proof. Expand the trace of $(R - \vartheta D)^p$ as noncommutative words and use cyclicity of the trace. Since D has rank one, $\operatorname{tr}(DR^{j-1}) = a_j$ and $\operatorname{tr}(DR^i DR^h) = a_{i+1}a_{h+1}$. For degree four the six words with two D's have traces $4a_1a_3 + 2a_2^2$; the four with three D's give $4a_1^2a_2$. The degree-two and degree-three counts give the remaining formulas. Differentiating at zero gives pa_p and spectral decomposition gives its eigenmode expression. The v-coordinate of $X\phi_1 = \lambda_1\phi_1$ proves the identity for h_v, and substitution gives the comparison. The tail bound uses $\sum_{i \geq 2} \phi_i(v)^2 = 1 - \phi_1(v)^2$.

Differentiating with $K_\vartheta = R - \vartheta D \succeq 0$ gives

$$\begin{aligned} Q_p'(\vartheta) &= p\operatorname{tr}(DK_\vartheta^{p-1}), \\ -Q_p''(\vartheta) &= p\sum_{i=0}^{p-2} \operatorname{tr}(DK_\vartheta^i DK_\vartheta^{p-2-i}). \end{aligned}$$

Write $D = a_1 ww^\top$ with $\|w\| = 1$. Each summand equals $a_1^2(w^\top K_\vartheta^i w)(w^\top K_\vartheta^{p-2-i} w)$, between zero and $a_1^2\|R\|^{p-2}$ since $0 \preceq K_\vartheta \preceq R$. Integrating twice gives the remainder bound. Finally $R - \vartheta D = (X + \beta e_v e_v^\top)^{-1}$ for $\beta = \vartheta/((1-\vartheta)r_1)$ when $\vartheta < 1$, by Sherman–Morrison. Thus $dz/d\beta = r_1$ at zero and the soft coefficient follows. □

Proposition A26 (A finite-removal stress case)**.** *Let $0 < t < 1/4$. On vertices $\{z, a, b, c\}$ put edges za, ab of weight t and zb, bc, cz of weight one. Ground the prescribed seed z and choose one further vertex. Apply the weighted version of the score in Zhou et al. [6], Eq. (4), $h_v = 2u_v\sum_{w \notin S} w_{vw}u_w$, using the exact smallest eigenvector. The score uniquely selects b, and its achieved smallest eigenvalue divided by the optimum is at most $2t$. For every real $p \geq 1$, exact minimization of $\operatorname{tr} X^{-p}$ over the three one-vertex choices uniquely selects the spectral optimum a.*

Proof. In the order a, b, c, the current grounded matrix is

$$X = \begin{pmatrix} 2t & -t & 0 \\ -t & 2+t & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

It is positive definite by grounded connectivity. Its off-diagonal graph is connected, so its smallest eigenvector is strictly positive: apply Perron–Frobenius to $dI - X$ with $d > \lambda_{\max}(X)$. The scores obey

$$h_a = 2tu_au_b > 0, \quad h_c = 2u_bu_c > 0, \quad h_b = h_a + h_c.$$

Consequently b is the unique maximizer, independently of how accurately the positive eigenvector is scaled. After selecting b

the matrix is $\mathrm{diag}(2t,2)$ and its smallest eigenvalue is $2t$. After selecting a, the matrix is

$$\begin{pmatrix} 2+t & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} + \mathrm{diag}(t,0) \succeq I,$$

so its smallest eigenvalue is at least one. After selecting c, the Rayleigh quotient at the coordinate a is $2t < 1$. This proves uniqueness of the spectral optimum and the ratio bound. The inverse-power trace after selecting a is at most two; after either alternative it is at least $(2t)^{-p} > 2$ for every $p \geq 1$. The inverse-power choice is therefore also uniquely a. □

The opposite spectral ordering also occurs within the same family. Consider the grounded matrix

$$X = \begin{pmatrix} 1 & -9/10 & 0 \\ -9/10 & 3 & -1/2 \\ 0 & -1/2 & 1 \end{pmatrix}.$$

Its three row sums are $1/10, 8/5, 1/2$, so it is realized by a connected weighted graph with one seed attached by those weights. The follower path and positivity imply that Zhou uniquely chooses the middle vertex. Its resulting matrix is I_2, giving spectral value one and inverse trace two. Deleting the first vertex gives eigenvalues $2 \pm \sqrt{5}/2$, inverse trace $16/11$, and inverse power traces $168/121, 1984/1331, 24352/14641$ for $p = 2,3,4$, each below two. Deleting the third leaves a smallest eigenvalue $2 - \sqrt{181}/10 < 7/10$, so every $p \geq 2$ inverse-power trace then exceeds two. Thus each of these three exact power-greedy choices improves CFCC but has spectral value $2 - \sqrt{5}/2 < 1$. The two explicit examples realize distinct orderings of the two metrics.

APPENDIX I
ANALYZED ORACLE AND SELECTION VARIANTS

This appendix develops the analyzed alternatives to the implemented CFCC hybrid: integer fresh-round selection, positive quadrature for noninteger powers, and permanently valid lazy bounds.

A. Fresh-round integer sketches

At each current SDDM matrix $A = C^\top C$, set $P = ZC$ for a new Gaussian Z, and draw an independent new Gaussian Q. Squared columns of $QA^{-\ell}$ and $PA^{-\ell}$ have targets $e_v^\top A^{-2\ell} e_v$ and $e_v^\top A^{-(2\ell-1)} e_v$. For a fixed unit vector w, the Gaussian concentration bound is

$$\mathbb{P}\{|\|Qw\|^2 - 1| > x\} \leq 2e^{-dx^2/8}, \quad 0 < x < 1. \tag{37}$$

It is the Gaussian case proved in Theorem A15. For approximate chains, the Frobenius error satisfies $E_\ell \leq (E_{\ell-1} + \rho_\ell)/a$, as proved in Proposition A17. Expanding the perturbed column-norm square converts this into moment error. The SDDM primitive used below solves to relative energy precision ξ in $\widetilde{O}((m_E + n)\log(1/\xi))$ work, with logarithmic failure-probability factors [12]. Here $m_E = m$ is the input edge count, upper-bounding every round's active off-diagonal count; diagonal entries contribute $O(n)$.

a) FreshRoundGreedy (all-node worst-case variant).: Fix a seed S_0, k additional vertices, integer $p \geq 1$, $0 < \varepsilon \leq 1/2$, and $0 < \delta < 1$. Write n for an upper bound on the active dimension and set

$$\eta = \varepsilon/2, \quad \zeta = \varepsilon/(32p), \quad J = p+1, \quad \ell_{\max} = \lceil J/2 \rceil, \\ d = \left\lceil 128\zeta^{-2} \log \frac{12nJk}{\delta} \right\rceil. \tag{38}$$

At each of the k rounds:

1) Form the current SDDM matrix A and its sparse factor $A = C^\top C$ using the restricted incidence factor. Draw entirely new independent Gaussian families Q, Z, with variance $1/d$ entries, and put $P = ZC$. These families and the solver randomness are independent of all previous rounds.
2) Apply repeated solves to these families through depth $\ell_{\max}$, obtaining every $r_{j,v} = e_v^\top A^{-j} e_v$, $1 \leq j \leq J$, with relative tolerance ζ. Specifically, squared columns of $QA^{-\ell}$ estimate order 2ℓ, and those of $PA^{-\ell}$ estimate order $2\ell - 1$. Discard orders above J.
3) Evaluate $\widehat{\Delta}_v = \widehat{q}_{p,v}(0)$ for every eligible vertex by the scalar logarithmic recurrence, select a maximizer, and ground it. Discard this round's sketches.

Each round's conditional failure budget is δ/k, including all its candidate estimates and all its solves.

Theorem A27 (Fresh-round all-node greedy guarantee)**.** *FreshRoundGreedy returns k additional vertices U_k with*

$$\Phi_p(S_0) - \Phi_p(S_0 \cup U_k) \\ \geq (1 - e^{-1} - \varepsilon) \max_{|U| \leq k} \big(\Phi_p(S_0) - \Phi_p(S_0 \cup U)\big) \tag{39}$$

with probability at least $1 - \delta$, for hard and soft grounding.

Proof. Condition on the complete history before round i. Its matrix, factor, and eligible vertices are now fixed. The new Gaussian families have the covariance and chi-squared laws stated above. There are at most nJ moment estimates; (37), with deviation $\zeta/4$, and (38) give conditional JL failure at most

$$2nJ \exp(-d\zeta^2/128) \leq \delta/(6k).$$

The residual/norm budgets in Theorem A28 give conditional failure at most $\delta/(3k)$ for each of the Gaussian norm and solver events. Their intersection implies relative moment error at most ζ, by the norm-perturbation bound above. In particular the total conditional failure is at most δ/k, for every history.

Let E_i be this round's simultaneous event. Taking conditional expectations and then a union bound gives

$$\mathbb{P}\left(\bigcup_{i=1}^{k} E_i^c\right) \leq \sum_{i=1}^{k} \mathbb{E}\big[\mathbb{P}(E_i^c \mid \mathscr{H}_{i-1})\big] \leq \delta.$$

Independence between the events E_i themselves is unnecessary; it is the fresh random draws conditional on the history that justify each probability bound.

On their intersection, Theorem A20 gives $(1-\eta)\Delta_v \le \widehat{\Delta}_v \le (1+\eta)\Delta_v$ for all eligible vertices in every round. If v maximizes the estimate and u maximizes the true marginal, then

$$\Delta_v \ge \frac{\widehat{\Delta}_v}{1+\eta} \ge \frac{1-\eta}{1+\eta}\Delta_u \ge (1-\varepsilon)\Delta_u.$$

The normalized seeded gain is monotone and submodular by Theorem A2. The same gap-closing induction as in Theorem A16 therefore gives the stronger factor $1-(1-(1-\varepsilon)/k)^k \ge 1-e^{-1}-\varepsilon$. For hard grounding the last allowed deletion, if any, uses the one-dimensional marginal and leaves zero trace; no inverse of an empty matrix is formed. □

Theorem A28 (Worst-case cost with explicit polynomial dependence)**.** *Suppose certified bounds $0 < a \le \lambda_{\min}(A)$ and $\lambda_{\max}(A) \le B$ hold for every round, and put $\kappa = B/a$. Using the nearly-linear SDDM primitive stated above, FreshRoundGreedy has*

$$T = \widetilde{O}\big(p^4 k(m_E+n)\varepsilon^{-2}\big). \tag{40}$$

Thus one explicit choice is $f(p) = p^4$; on connected graphs $m_E + n = O(m_E)$ gives the $\widetilde{O}(f(p)km_E\varepsilon^{-2})$. The suppressed factors are logarithmic in $n,k,p,\kappa,1/\varepsilon,1/\delta$, with no suppressed polynomial factor in p. The count of SDDM right-hand-side solves alone is $O(kdL) = \widetilde{O}(p^3k\varepsilon^{-2})$. These are input-wise worst-case bounds, independent of the observed CELF refresh count, in the standard capped high-probability solver model; solver failure is included in δ.

Proof. Normalize every matrix by the common B, and also replace the soft penalty by β/B. All original marginal values acquire the same positive factor B^{-p} on converting back, so every selection is unchanged. The normalized spectrum is in $[1/\kappa, 1]$. Its factor satisfies $\mathbb{E}\|P\|_F^2 = \operatorname{tr} A \le n$ and $\mathbb{E}\|Q\|_F^2 \le n$. Set

$$b_{\mathrm{rnd}} = \sqrt{6nk/\delta}, \qquad \xi = \frac{\zeta}{64\ell_{\max}\kappa^{2\ell_{\max}}(b_{\mathrm{rnd}}+1)}. \tag{41}$$

Markov's inequality gives simultaneous starting norms at most b_{rnd} except with conditional probability $\delta/(3k)$. Use relative energy accuracy ξ at every inverse-power solve, with per-solve conditional failure at most $\delta/(6kdL)$. There are at most $2d\ell_{\max}$ such solves per round. The norm test and a capped solve budget may terminate with an arbitrary feasible answer on failure, so the work bound also holds on unsuccessful executions.

For either family, let V_0 be its starting batch and $\widehat{V}_j$ its approximate chain. Write $\rho_j = \|\widehat{V}_j A - \widehat{V}_{j-1}\|_F$ and $E_j = \|\widehat{V}_j - V_0 A^{-j}\|_F$. Relative energy accuracy implies $\rho_j \le \xi\sqrt{\kappa}\|\widehat{V}_{j-1}\|_F$. Inductively $\|\widehat{V}_{j-1}\|_F \le \kappa^{j-1}(b_{\mathrm{rnd}}+1)$, whence for $j \le \ell_{\max}$

$$\rho_j \le \frac{\zeta}{64\ell_{\max}\kappa^{\ell_{\max}+1/2}} \le \frac{\zeta}{64\ell_{\max}\kappa^{\ell_{\max}}},$$
$$E_j \le \kappa(E_{j-1}+\rho_j) \le \zeta/64.$$

The last inequality follows by summing at most $\ell_{\max}$ terms, each amplified by at most $\kappa^{\ell_{\max}}$, starting with $E_0 = 0$. It also closes the stated induction on approximate batch norms. Every normalized target moment is at least one. Thus these absolute column errors, combined with the relative $\zeta/4$ JL event, meet the sufficient $\zeta/8$ norm budgets and give relative moment error at most ζ.

There are at most $2d\ell_{\max}$ solves per round, each on a matrix with $O(m_E+n)$ nonzeros. Sparse Gaussian-factor multiplication and moment accumulation cost $O(d(m_E+nL))$ per round. The scalar recurrences cost $O(np^2)$ and the maximization costs $O(n)$. The precision satisfies

$$\log(1/\xi) = 2\ell_{\max}\log\kappa + \log\big(64\ell_{\max}(b_{\mathrm{rnd}}+1)/\zeta\big).$$

A nearly-linear SDDM solve costs $\widetilde{O}((m_E+n)\log(1/\xi))$, including logarithmic failure-probability factors. Since $d = \widetilde{O}(p^2\varepsilon^{-2})$ and $\ell_{\max} \le p$, explicitly counting this precision cost gives

$$\widetilde{O}\big(k(m_E+n)p^2\ell_{\max}^2\varepsilon^{-2}\big) + O(knp^2) \subseteq \widetilde{O}\big(p^4k(m_E+n)\varepsilon^{-2}\big).$$

The p^3 solve count and the extra factor from solve precision are therefore distinguished. The argument applies to every input and every adaptive matrix sequence on the simultaneous success event. □

B. Positive scalar oracles and quadrature

Use the active matrix and penalty notation of Lemma A11. Higher inverse powers of shifted matrices give a positive representation for every exponent.

Proposition A29 (Positive scalar representation for all powers)**.** *Define, for integers $m \ge 1$,*

$$q_m(t) = \frac{(-1)^{m-1}}{(m-1)!}\frac{d^{m-1}}{dt^{m-1}}q_1(t).$$

Then $q_m(t) = \operatorname{tr}(A+tI)^{-m} - \operatorname{tr}(B+tI)^{-m} \ge 0$. For any $s > 0$ and any integer $m > s$,

$$\Delta_s = \frac{\Gamma(m)}{\Gamma(s)\Gamma(m-s)}\int_0^\infty t^{m-s-1}q_m(t)\,dt. \tag{42}$$

For integer $s = m$ one can instead use $\Delta_m = q_m(0)$. Each $q_m(t)$ is determined by the scalar moments $r_1(t),\dots,r_{m+1}(t)$ from the original matrix A.

Proof. Repeated differentiation of the shifted inverse gives $d^{m-1}(A+tI)^{-1}/dt^{m-1} = (-1)^{m-1}(m-1)!(A+tI)^{-m}$, which proves the trace identity. For a soft update the ordered eigenvalues of A_β are at least those of A, by the min–max principle. For a hard update, writing $d = \dim A$, interlacing gives $\lambda_i(A) \le \lambda_i(C)$ for $1 \le i \le d-1$. Because $x \mapsto (x+t)^{-m}$ decreases, in the soft case every matched eigenvalue contribution is nonnegative; in the hard case this also leaves the positive contribution of $\lambda_d(A)$. This proves $q_m \ge 0$.

The substitution $t = xu$ in the beta integral gives

$$\int_0^\infty t^{m-s-1}(x+t)^{-m}dt = x^{-s}\frac{\Gamma(m-s)\Gamma(s)}{\Gamma(m)}.$$

Apply it to the finite eigenvalue sums and subtract. The integral of the difference is finite: near zero its integrand is bounded by a constant times t^{m-s-1}, and near infinity by a constant times

t^{-s-1}. This proves (42) as a positive representation, including $s \geq 1$.

Here is an explicit scalar evaluation rule. Put

$$F(t) = \begin{cases} 1+\beta r_1(t), & \beta < \infty, \\ r_1(t), & \beta = \infty. \end{cases} \qquad q_1(t) = -\frac{d}{dt}\log F(t).$$

The latter identity follows immediately from (22) and $r_1' = -r_2$. In a neighborhood of $z = 0$,

$$r_1(t+z) = \sum_{j\geq 0} (-1)^j r_{j+1}(t) z^j,$$
$$F(t+z) = \sum_{j\geq 0} f_j z^j,$$
$$\log F(t+z) = g_0 + \sum_{j\geq 1} g_j z^j.$$

The resolvent series converges for $|z| < \lambda_{\min}(A) + t$; $F(t) > 0$ also ensures a local logarithm. For soft grounding $f_0 = 1 + \beta r_1$ and $f_j = \beta(-1)^j r_{j+1}$ for $j \geq 1$; for hard grounding $f_j = (-1)^j r_{j+1}$ for all $j \geq 0$. Equating coefficients in $F' = F(\log F)'$ gives, successively,

$$g_k = \frac{k f_k - \sum_{j=1}^{k-1} j g_j f_{k-j}}{k f_0}, \qquad q_m(t) = (-1)^m m g_m. \tag{43}$$

Thus moments through order $m+1$ suffice. For example,

$$q_2(t) = \begin{cases} \dfrac{2\beta r_3}{1+\beta r_1} - \dfrac{\beta^2 r_2^2}{(1+\beta r_1)^2}, & \beta < \infty, \\ \dfrac{2r_3}{r_1} - \dfrac{r_2^2}{r_1^2}, & \beta = \infty. \end{cases}$$

Its nonnegativity, and that of every higher-order expression, follows from the trace/interlacing argument above. □

For the quadrature proof, let $N_A(x)$ and $N_B(x)$ count active eigenvalues at most x, and put $\xi(x) = N_A(x) - N_B(x)$. The rank-one and principal-submatrix interlacing inequalities give $0 \leq \xi \leq 1$. If both spectra lie in $[a, B_*]$, integration of each scalar eigenvalue contribution gives $\Delta_f = \int_a^{B_*} (-f')\xi$ for soft grounding and $\Delta_f = f(B_*) + \int_a^{B_*} (-f')\xi$ for hard grounding. The next statement calls the spectral upper bound B.

Theorem A30 (Fixed positive surrogate with logarithmic shift count)**.** *Suppose every nonempty admissible grounded matrix has spectrum in $[a,B]$, $a > 0$, and put $\kappa = B/a$. For each $s > 0$ and $0 < \eta < 1$, there is a fixed positive rule*

$$\widetilde{f}_s(x) = \sum_{j=1}^{q} w_j (x + t_j)^{-m}, \quad t_j, w_j > 0, \quad m = \lfloor s \rfloor + 1,$$

such that its trace objective is nonincreasing and supermodular at every rule order and, for every soft or hard marginal,

$$(1-\eta)\Delta_{x^{-s}} \leq \Delta_{\widetilde{f}_s} \leq \Delta_{x^{-s}}. \tag{44}$$

One explicit choice is

$$p = m + 3 + \lceil \log_4(2/\eta) \rceil, \qquad q \leq (\lceil \log_2 \kappa \rceil + 2)p.$$

Thus $q = O_s((1+\log\kappa)\log(1/\eta))$ for a fixed exponent. For $0 < s < 1$ the rule uses only first inverse powers ($m = 1$); for $s = 1$ one can use the exact single shift $t = 0$ instead.

Proof. Use the positive integral in Proposition A29. Partition its shift domain into $[0,a]$, geometric intervals $[a,2a],[2a,4a],\ldots$ truncated at B, and $[B,\infty)$. On $[0,a]$ use $t = az$ and p-node Gauss–Jacobi for the weight z^{m-s-1} on $[0,1]$. On every interior interval use p-node Gauss–Legendre, including the positive factor t^{m-s-1} in the sampled integrand. On the tail use $z = B/t$ and Gauss–Jacobi with weight z^{s-1}. The tail integrand becomes $B^{-s}(1+xz/B)^{-m}$, a positive multiple of $(x+B/z)^{-m}$. All nodes and coefficients are positive and independent of S or the candidate. Write the resulting scalar approximation as $F_p(x)$.

Here is a uniform error bound for both f_s and $g_s = -f_s'$. Put $\alpha = s+1-m \in [0,1)$ and $D = 2p-1$. For an interior interval with center c write $y = (t-c)/c$; its doubling ratio ensures $|y| \leq 1/3$. After extracting a positive factor, the integrands for the function and its negative derivative have the form $(1+y)^{-\alpha}(1+\theta y)^{-r}$ with $0 \leq \theta \leq 1$ and $r = m, m+1$. On $|y| = 2/3$ their modulus is at most $3^{\alpha+r}$, while their minimum on the real interval is at least $(4/3)^{-\alpha-r}$. Cauchy's coefficient bound shows that the Taylor tail beyond degree D is at most $2\,3^{\alpha+r}2^{-(D+1)}$. The quadrature integrates the Taylor polynomial exactly; positivity and exactness on constants bound its error plus the integral's error by twice this remainder times the interval mass. Dividing by the true integral gives relative error at most $4\,4^{\alpha+r}4^{-p} \leq 4^{m+3-p}$.

For the lower endpoint the weight absorbs the singularity, leaving $(x+az)^{-r}$. Expanding about $z = 1/2$ gives a normalized variable of radius $a/(2x+a) \leq 1/3$ and the same, or smaller, bound. For the upper endpoint the derivative integrand, under its fixed weight, is proportional to $z(1+xz/B)^{-m-1}$. Approximate its denominator factor to degree $D-1$ and multiply by z. The error argument now uses exactness on z, with radius $x/(2B+x) \leq 1/3$. The bound is $8\,4^{m+1}4^{-p} \leq 4^{m+3-p}$. The function tail needs degree D and has a smaller bound. These estimates hold for every $x \in [a,B]$. Summing positive panel integrals gives, with $\varepsilon = 4^{m+3-p} \leq \eta/2$,

$$|F_p/f_s - 1| \leq \varepsilon, \qquad |(-F_p')/g_s - 1| \leq \varepsilon.$$

Define $\widetilde{f}_s = F_p/(1+\eta/2)$. Both it and its negative derivative lie between $1-\eta$ and 1 times their true counterparts.

For soft grounding integrate the derivative bounds against the nonnegative spectral count ξ. For hard grounding use $\Delta_f = f(B) + \int_a^B (-f')\xi$, so the endpoint value is controlled as well. This proves (44) without taking a relative error of two nearly equal traces. In particular it treats the hard ratio r_2/r_1 rather than a log-resolvent substitute. Every shifted inverse power is CM, so Theorem A2 proves supermodularity of every finite positive rule, including orders below the accuracy threshold. There are at most $\lceil \log_2 \kappa \rceil + 2$ panels, proving the count. □

C. All-node shifts and certified lazy evaluation

For a fixed shifted system $R_t = (A+tI)^{-1}$, take $P = ZC$, Q independent, and reuse these right-hand sides across shifts. The

squared columns of QR_t estimate the second resolvent moment; those of $(P+\sqrt{t}Q)R_t$ estimate the first, since $R_t(A+tI)R_t = R_t$. This finite-shift construction avoids an additional integral for the first moment. For a common right-hand side, unpreconditioned shift-invariant Krylov spaces provide another implementation of the same shifted solves. Its work retains the Krylov depth explicitly.

For deterministic refinement, a single Lanczos run yields Gauss lower and left-Radau upper bilinear bounds for the CM function $-f'$. Integrating these bounds along the grounding penalty gives marginal intervals. Their one-sided signs follow from the even and odd Hermite remainders, respectively; finite hard-tail bounds control the infinite penalty endpoint.

A certified lazy rule starts with valid marginal upper keys. At the current heap top v, refine an interval $L_v \le \Delta_v \le H_v$ until $L_v \ge (1-\theta)H_v$. Replace its key by the minimum of the old and fresh upper bounds; accept if $L_v \ge (1-\theta)U_{\text{next}}$, otherwise reinsert it. Supermodularity preserves all old upper bounds. A rejected fresh candidate falls strictly below another key; when it reaches the top again in the same round, its relative-width condition makes it pass. Thus each candidate is refreshed at most once per round. Every accepted marginal is at least $(1-\theta)$ of the current maximum. The gap induction in Theorem A16 gives $1-(1-(1-\theta)/k)^k \ge 1-e^{-1}-\theta$ for seeded gain. The same induction with a fixed size-k comparison set proves the augmented-budget statement in Corollary A24.

For general heat/CM refinement, the required Lanczos depth also depends on the chosen function and, for heat, its horizon. The integer fresh-round cost and the CFCC hybrid accounting use their respective stated oracle models.

APPENDIX J
IMPLEMENTATION DETAILS

The engine stores both solution sketches transposed, as $N \times d$ arrays with unscaled ± 1 entries generated by a seeded hash of the row index and the edge (for Π_C) or vertex (for Π_I) index, and divides squared norms by d; this is equivalent to the variance-$1/d$ convention of (5). A candidate is marked *refreshed* once its moments have been computed by numerical refresh solves; the analysis assigns those solves the explicit error budgets of Appendix E. The round heap orders equal keys by decreasing vertex index, and only a strictly larger refreshed value replaces the incumbent. A reused refresh column keeps its refresh tolerance ε_{ref}. The approximate-Cholesky factor is applied to blocks of at least four right-hand sides, and Jacobi-preconditioned CG is used for narrower blocks; on roadNet-PA the factor is applied to every block, including single right-hand sides. The sketch-only ablation keys every candidate from the solution sketches in every round, and the scalar-cache ablation keeps no solution sketches.

APPENDIX K
ADDITIONAL EXPERIMENTAL DETAILS

The shared seed stage of the main comparison is listed in Table XI. These costs are common to all compared selection methods. Adding the same seed cost to both methods preserves their ordering while changing an end-to-end speed ratio.

TABLE XI
SHARED JL SEED-STAGE TIMES, DIMENSION 128.

Graph	Seed time (seconds)
DBLP	3.747
Amazon	5.400
YouTube	12.710
web-Google	12.972
Skitter	21.157
roadNet-PA	25.343
LiveJournal	62.708